



SOCIETY FOR ECOSYSTEM RESTORATION
IN NORTHERN BRITISH COLUMBIA

Restoring Fish Passage in the Skeena Region - 2025

Prepared for
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Ministry of Transportation and Infrastructure

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on behalf of
Society for Ecosystem Restoration in Northern British Columbia

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Claude Opus 5 (Anthropic) assisted with literature synthesis, drafting, and technical writing. All scientific interpretation, data analysis, and conclusions are the responsibility of the authors.



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Acknowledgement

At New Graph Environment, we understand our well-being as inseparable from the health of the land and waters we work within. When we care for ecosystems, we care for ourselves and for the communities connected to them.

Modern civilization has a long journey ahead to acknowledge and address the historic and ongoing impacts of colonialism that have resulted in harm to the cultures and livelihoods living interconnected with our ecosystems for many thousands of years.

Executive Summary

 [Executive Summary \(PDF\)](#)

This report is available as a PDF and as an online interactive report at https://www.newgraphenvironment.com/fish_passage_skeena_2025_reporting/. We recommend viewing online as the web-hosted HTML version contains more features and is more easily navigable. Please reference the website for the latest version stamped PDF from [fish_passage_skeena_2025_reporting.pdf](#).

Since 2020, the Society for Ecosystem Restoration Northern British Columbia (SERNbc) has been actively involved in planning, coordinating, and conducting fish passage restoration efforts within the Bulkley River and Morice River watershed groups, which are sub-basins of the Skeena River watershed. In 2022, the study area was expanded to include the Zymoetz River watershed group and the Kispiox River watershed group, followed by an extension in 2023 to encompass sections of the Kitsumkalum River watershed group, particularly where Highway 16 intersects the watershed.

The primary objective of this project is to identify and prioritize fish passage barriers within study areas, develop comprehensive restoration plans to address these barriers, and foster momentum for broader ecosystem restoration initiatives. While the primary focus is on fish passage, this work also serves as a lens through which to view the broader ecosystems, leveraging efforts to build capacity for ecosystem restoration and improving our understanding of watershed health. We recognize that the health of life - such as our own - and the health of our surroundings are interconnected, with our overall well-being dependent on the health of our environment.

Although the main purpose of this report is to document 2025 field work data and results, it also builds on reporting from past field activities and all reports can be considered living documents that can be updated and improved over time.

- [Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning \(2020\)](#)
- [Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning 2021](#)
- [Bulkley River Watershed Fish Passage Restoration Planning 2022](#)
- [Skeena Watershed Fish Passage Restoration Planning 2022](#)
- [Skeena Watershed Fish Passage Restoration Planning 2023](#)
- [Skeena Watershed Fish Passage Restoration Planning 2024](#)

Fish passage assessment procedures conducted through SERNbc in the Bulkley River, Kispiox River, Kalum River, Morice River, and Zymoetz River watershed groups since 2020 are amalgamated online within the Assessment Data Summary appendix of the report found [here](#) which includes links to project reporting for each site.

Executive Summary

The 2025 field season was directed at effectiveness monitoring and environmental DNA sampling rather than at new fish passage assessments. Two crossings already inventoried in PSCIS were reassessed; no Phase 1 assessments were completed.

No Phase 2 habitat confirmation assessments were conducted. Habitat confirmations follow from crossings prioritised during Phase 1 assessment, and no Phase 1 work was completed this season.

Environmental DNA (eDNA) sampling was the substantive field programme this season. A total of 19 samples were collected across 10 streams, alongside 1 field and office blanks run as protocol controls.

Monitoring was conducted at two sites, including post-remediation evaluations at Waterfall Creek (PSCIS crossing 124421) and Peacock Creek (PSCIS crossing 197962). Monitoring included eDNA sampling and habitat observations. A custom effectiveness monitoring form tailored to fish passage projects was used and metrics assessed included flow velocity, substrate condition, channel constriction, riparian condition, and cover availability.

A major challenge in advancing fish passage restoration is the complexity of working across jurisdictions and with multiple stakeholders—rail and highway authorities, forestry ministries, licensees, and private landowners. These partners are often being asked to accommodate priorities that originate outside their mandates and budgets. Convincing them to invest in difficult, high-cost interventions—like modifying crossings or relocating infrastructure—requires navigating uncertainty about costs and ecological outcomes, as well as a disconnect between the benefits to watershed health and the internal pressures or performance goals of these agencies. It's a tough ask: to take on massive, uncertain projects when they're already stretched thin with their own responsibilities.

Fish passage restoration across British Columbia is further complicated by the legacy of infrastructure deeply embedded in the landscape. Roads, railways, highways, community infrastructure and private assets often constrain floodplains and disrupt natural hydrological processes. While targeted repairs to individual barriers are essential, they won't resolve the broader systemic issues without rethinking and restructuring how infrastructure interacts with watershed function. Loss of riparian vegetation and intensive beaver management only add to the degradation. Addressing these challenges means making strategic, well-communicated choices, building trust, and staying committed to a longer-term transformation.

A single definitive ranking of remediation priorities is not achievable, because the criteria that determine whether a crossing is worth remediating are not measured in common units, and the

deciding constraint sits outside the project in the capacity and willingness of infrastructure owners and tenure holders to support implementation across multi-year timelines. What can be done systematically is to maintain a well-characterised set of candidate sites, so that when a tenure holder is in a position to act, the ecological value of the opportunity and the appropriate scope of work are already understood.

Looking ahead, priorities for the program are:

- Continue and extend the long-term effectiveness monitoring program so decisions elsewhere are better informed.
- Continue environmental DNA sampling as an annual, region-wide layer, building single-year detections into distribution and trend information.
- Extend functional floodplain mapping and the climate departure analysis across the project area, so watershed condition sits alongside barrier status wherever prioritization is done.
- Commission engineering designs at the Ministry of Transportation and Infrastructure crossings identified in previous reporting, and advance the highest ranked sites assessed this year.
- Resolve species use through targeted fish sampling where habitat ranks high but detections are absent or unresolved.
- Continue working with First Nations, stewardship groups, educational institutions and tenure holders, feeding monitoring results back into prioritization and design as adaptive management.

1 Introduction

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This project builds on Society for Ecosystem Restoration Northern BC (SERNbc) work in:

- [Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning \(2020\)](#) (Irvine 2021)
- [Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning 2021](#) (Irvine [2021] 2022)
- [Bulkley River Watershed Fish Passage Restoration Planning 2022](#) (irvine_etal2023BulkleyWatershed?)
- [Skeena Watershed Fish Passage Restoration Planning 2022](#) (irvine_wintersheidt2023SkeenaWatershed?)
- [Skeena Watershed Fish Passage Restoration Planning 2023](#) (irvine_schick2023SkeenaWatershed?)
- [Skeena Watershed Fish Passage Restoration Planning 2024](#) (irvine_schick2025SkeenaWatershed?)

The health and viability of freshwater fish populations can depend on access to tributary and off channel areas which provide refuge during high flows, opportunities for foraging, overwintering habitat, spawning habitat and summer rearing habitat (Bramblett et al. 2002; Swales and Levings 1989; Diebel et al. 2015). Culverts can present barriers to fish migration due to low water depth,

1 Introduction

increased water velocity, turbulence, a vertical drop at the culvert outlet and/or maintenance issues (Slaney and Zaldokas 1997; Cote et al. 2005). As road crossing structures are commonly upgraded or removed there are numerous opportunities to restore connectivity by ensuring that fish passage considerations are incorporated into repair, replacement, relocation and deactivation designs.

Although remediation and replacement of stream crossing structures can have benefits to local fish populations, the costs of remedial works can be significant and the impacts of the work often complex to evaluate and quantify. Additionally, allocation of ecosystem restoration funding towards infrastructure upgrades on transportation right of ways are not always considered ethical under all circumstances from all perspectives. When funds are finite and invested groups are engaged in fund raising, cost benefits and the ethics of crossing replacements should be explored collaboratively alongside the cost benefits and ethics of alternative investment activities including transportation corridor relocation/deactivation, land procurement/covenant, cattle exclusion, riparian/floodplain restoration, habitat complexing, water conservation, commercial/recreational fishing management, salt water interventions and research.

Please note that at the time of reporting this document can be considered a living document. See the [Open Source Reporting Framework \(page 35\)](#) section for details on version tracking and project documentation.

2 Background

2.1 Approach

Fish passage assessments in British Columbia follow the provincial protocol (MoE 2011), which scores barrier risk at road-stream crossings from structural parameters — culvert embedment, slope, outlet drop, and culvert diameter relative to channel width. The “Barrier” and “Potential Barrier” classifications represent passability for juvenile salmon or small resident rainbow trout under any flow conditions that may occur throughout the year (Clarkin et al. 2005; Bell 1991; Thompson 2013). The thresholds are deliberately conservative, so the classifications are risk scores rather than determinations of whether fish actually move through. Many structures classified as “Barrier” still pass some species and life stages at some flows, and natural features in adjacent reaches can be as limiting as the structure itself. Passability can be quantified in other ways — fish length, swim speed, and species physiology all affect connectivity outcomes — and finer-grained scoring methods exist for structures already classified as barriers (Bourne et al. 2011; Kemp and O’Hanley 2010; Washington Department of Fish & Wildlife 2009). Whether a remediation changes outcomes for fish therefore depends on the habitat upstream, the species and life stages in question, and the broader condition of the reach the structure connects to — not the classification alone.

The provincial inventory contains thousands of structures on fish-bearing streams. Replacement at a typical forest road culvert is in the six-figure range; structural replacements on highway and rail corridors run into seven and eight figures. Funding for restoration is finite and split across multiple programs and partners. Prioritization across the inventory therefore carries more weight than any individual assessment — the assessment establishes what could be fixed, while prioritization decides what should be.

Authority and funding for these structures are distributed across rail authorities, highway ministries, forestry licensees, and private landowners. Each operates under its own mandate, performance metrics, and budget cycle. Advancing restoration involves working inside organizations whose primary mandate is moving goods, maintaining roads, or harvesting timber — not aquatic ecological outcomes. The work proceeds through shared data, sustained relationships, and incremental progress across multiple budget years and field seasons.

Many road and rail corridors were built without consideration of fish passage or floodplain function. Damage frequently extends beyond the structure at the crossing — channelized streams, dykes that sever lateral connection, drained wetlands, cleared riparian buffers. Replacing a culvert in-kind can restore longitudinal passage while leaving the broader corridor-scale disconnection unaddressed. Understanding where infrastructure constrains watershed function helps distinguish cases where structural replacement is sufficient from cases where remediation needs to address more than the structure itself.

2 Background

This report compiles the layers used for prioritization in the watershed groups where the work is active: structural assessment data, habitat modelling, climate trajectory, and floodplain extent. Each layer contributes a piece of the picture — what is present, what it could support, how it is changing, and the surrounding watershed context. The work happens over many years across many partners; this report documents one year's contribution to that effort.

2.2 Project Location

The study area includes the Bulkley River, Kispiox River, Kalum River, Morice River, and Zymoetz River watershed groups (Figure 2.1) and is within the traditional territories of the Wet'suwet'en, Gitxsan and Tsimshian.

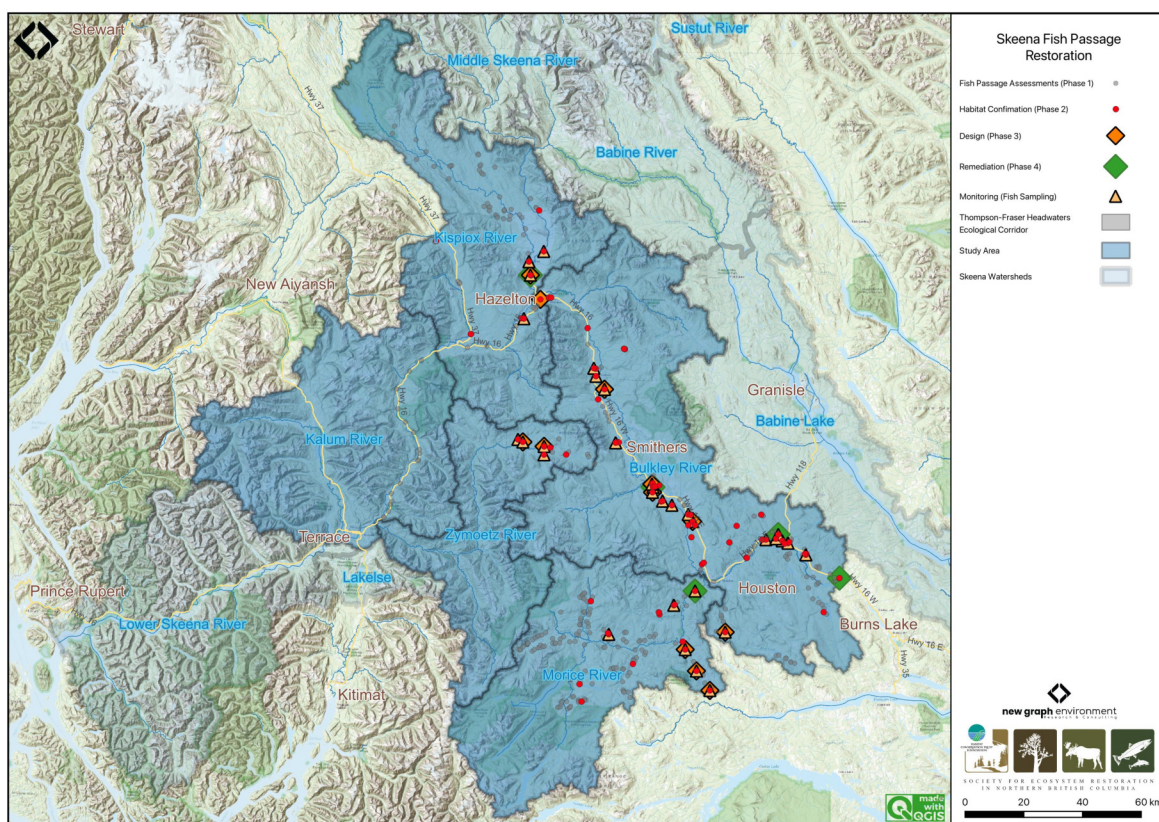


Figure 2.1: Overview map of the Skeena Region study areas

2.3 Wet'suwet'en

Wet'suwet'en hereditary territory covers an area of 22,000km² including the Bulkley River and Morice River watersheds and portions of the Nechako River watershed. The Wet'suwet'en people are a matrilineal society organized into the Gilseyhu (Big Frog), Laksilyu (Small Frog), Tsayu (Beaver clan), Gitdumden (Wolf/Bear) and Laksamshu (Fireweed) clans. Within each of the clans there are a number of kin-based groups known as Yikhs or House groups. The Yikh is a

2.4 Gitxsan

partnership between the people and the territory. Thirteen Yikhs with Hereditary Chiefs manage a total of 38 distinct territories upon which they have jurisdiction. Within a clan, the head Chief is entrusted with the stewardship of the House territory to ensure the Land is managed in a sustainable manner. Inuk Nu'at'en (Wet'suwet'en law) governing the harvesting of fish within their lands are based on values founded on thousands of years of social, subsistence and environmental dynamics. The Yintahk (Land) is the centre of life as well as culture and its management is intended to provide security for sustaining salmon, wildlife, and natural foods to ensure the health and well-being of the Wet'suwet'en (Office of the Wet'suwet'en 2013; "Office of the Wet'suwet'en" 2021; FLNRORD 2017).

2.4 Gitxsan

Gitxsan means "People of the River Mist". The Gitxsan Laxyip (traditional territories) covers an area of 33,000km² within the Skeena River and Nass River watersheds. The Laxyip is governed by 60 Simgiigyet (Hereditary Chiefs), within the traditional hereditary system made up of Wilps (House groups). Anaat are fisheries tenures found throughout the Laxyip. Traditional governance within a matrilineal society operates under the principles of Ayookw (Gitxsan law) ("Gitxsan Huwilp Government," n.d.). Many band members live in Hazelton, Kispiox and Glen Vowell (the Eastern Gitxsan) as well as within Kitwanga, Kitwankool and Kitsegukla (the Western Gitxsan) (Powell et al. 2018).

2.5 Tsimshian

The Kitsumkalum community, part of the Tsimshian Nation, maintains a rich cultural heritage rooted in ancient traditions and values. Their society, governed by Tsimshian Law (ayaawx), emphasizes strong connections through marriages, adoptions, and resource sharing with other Tsimshian tribes. The community upholds its cultural and spiritual practices, including fishing, harvesting, and land stewardship, despite the impacts of colonization (Kitsumkalum Band, n.d.).

Kitsumkalum's social structure is based on matrilineal kinship, with significant emphasis on family ties through the mother's lineage. Their cultural identity is expressed through crest groups (pteex), lineage houses (waap), and the importance of landed property (laxyuup), which ties them to their ancestral territories. The community combines traditional governance with modern administrative functions, reflecting their resilience and commitment to preserving their heritage (Kitsumkalum Band, n.d.).

The Kitsumkalum River salmon populations have been an important part of their culture and economy (Gottesfeld and Rabnett 2007).

2.6 Bulkley River

The colonial naming conventions for the Bulkley and Morice rivers can be misleading. The Neexdzii Kwah (Upper Bulkley River) is a tributary to Widzin Kwah (Morice River), contributing on average less than one-third of the total flow at their confluence in Houston, BC. However, historical naming during early mapping and settlement resulted in “Bulkley River” being applied to the mainstem flowing from Houston to its confluence with the Skeena River at Hazelton. From a hydrological perspective, it may be more accurate to consider the Morice River as the mainstem and the Upper Bulkley River as one of its tributaries. These naming differences can obscure the true flow dynamics of the watershed. By using Wet’suwet’en place names, we not only recognize the deep connection the Wet’suwet’en have to their clan territories but also convey a more accurate understanding of the geography and hydrology of the region.

The Bulkley River (Neexdzii Kwah), as defined by colonial naming, is an 8th order stream that drains an area of 7,762km² in a generally northerly direction from Bulkley Lake on the Nechako Plateau to its confluence with the Skeena River at Hazelton. It has a mean annual discharge of 139.2m³/s at station 08EE004 near Quick (~27km south of Telkwa), and 18m³/s at station 08EE003 upstream near Houston. Flow at Quick is heavily influenced by the Morice River (Widzin Kwah), which joins the Bulkley just downstream of Houston. As a result, discharge patterns at Quick are characteristic of high-elevation, snow-dominated watersheds, with peak flows typically occurring from May to July (Figure [2.2](#)). Upstream of Houston, the hydrograph peaks earlier (May–June), consistent with lower-elevation terrain and reduced snowpack influence (Figure [2.3](#)).

Changes to the climate systems are causing impacts to natural and human systems on all continents with alterations to hydrological systems caused by changing precipitation or melting snow and ice increasing the frequency and magnitude of extreme events such as floods and droughts (Calvin et al. 2023; ECCC 2016). These changes are resulting in modifications to the quantity and quality of water resources throughout British Columbia and are likely to compound issues related to drought and flooding in the Bulkley River watershed where numerous water licenses are held with a potential over-allocation of flows identified during low flow periods (ILMB 2007).

The valley bottom has seen extensive settlement over the past hundred years with major population centers including the Village of Hazelton, the Town of Smithers, the Village of Telkwa and the District Municipality of Houston. As a major access corridor to northwestern British Columbia, Highway 16 and the Canadian National Railway are major linear developments that run along the Bulkley River within and adjacent to the floodplain with numerous crossing structures impeding fish access into and potentially out from important fish habitats. Additionally, as the valley bottom contains some of the most productive land in the area, there has been extensive conversion of riparian ecosystems to hayfields and pastures leading to alterations in flow regimes, increases in water temperatures, reduced streambank stability, loss of overstream cover and channelization (ILMB 2007; Wilson and Rabnett 2007).

2.6 Bulkley River

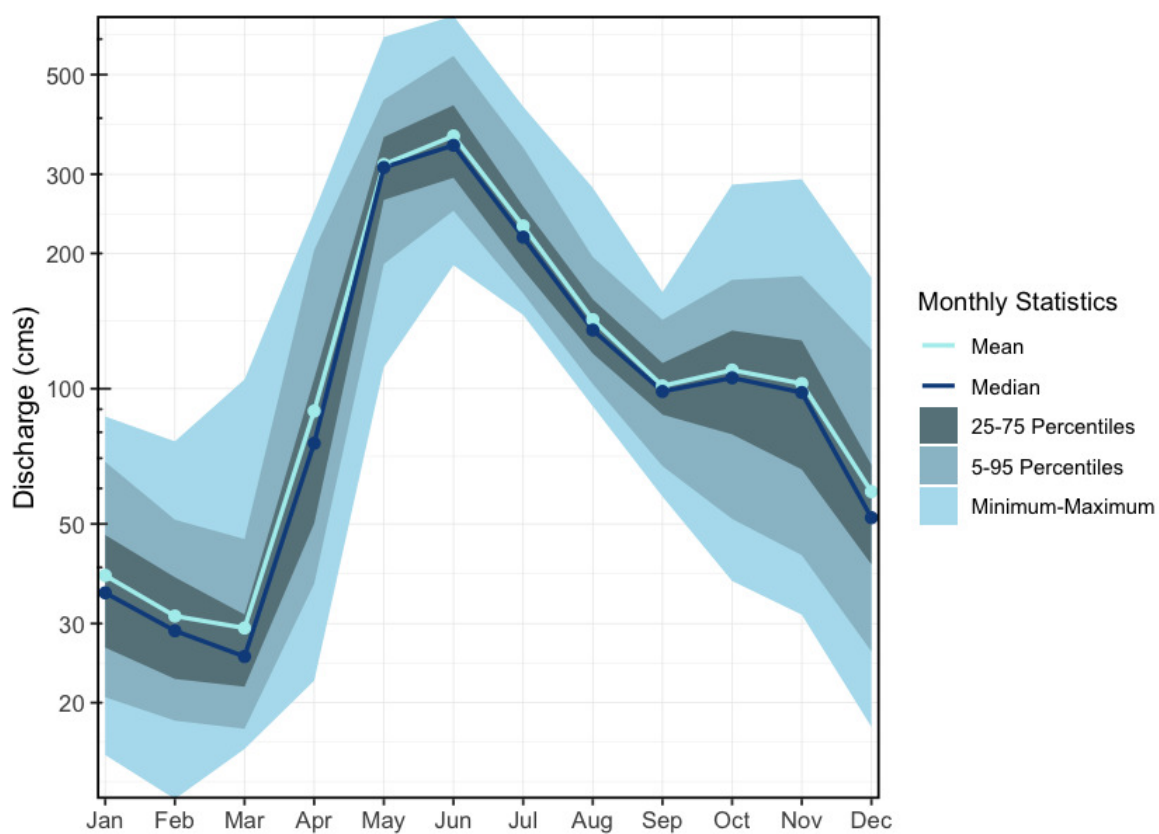


Figure 2.2: Hydrograph for Bulkley River at Quick (Station #08EE004). Available mean daily discharge data from 1930 to 2023.

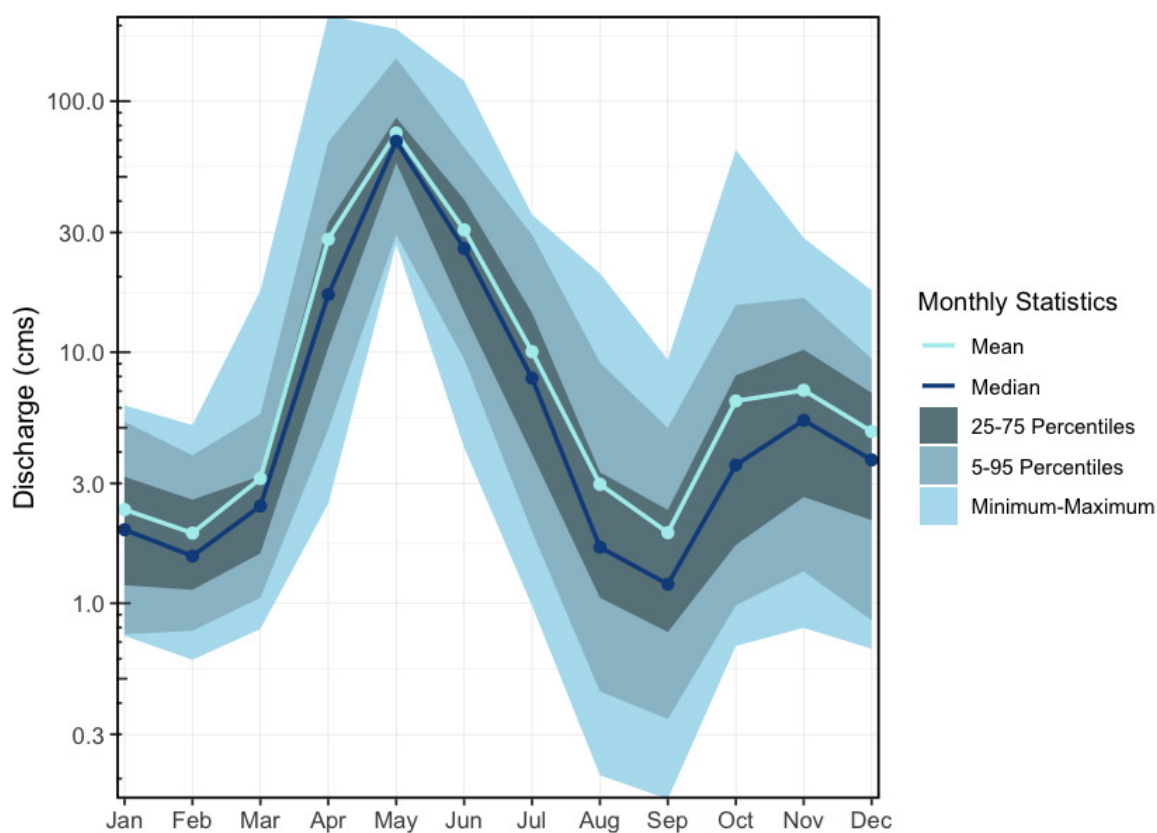


Figure 2.3: Hydrograph for Upper Bulkley River near Houston, upstream of the Morice River confluence (Station #08EE003). Available mean daily discharge data from 1930 to 2023.

2.7 Morice River

The Morice River, known to the Wet'suwet'en as Widzin Kwah and meaning river of clear, blue-green waters, flows from its headwaters at Morice Lake (Widzin Bin) northeast to its confluence with the Upper Bulkley River (Neexdzii Kwah) just north of Houston, BC. The river is an 8th order stream that drains 4,379km² of the Coast Mountains and Interior Plateau. Major tributaries include the Nanika, Atna, and Thautil rivers, as well as Gosnell Creek. Numerous large lakes are located in the southern portion of the watershed, including Morice Lake, McBride Lake, Stepp Lake, Nanika Lake, Kid Price Lake, and Owen Lake. One active hydrometric station (08ED002), located near the outlet of Morice Lake, records a mean annual discharge of 75m³/s (Canada 2024). Flow patterns are characteristic of high-elevation coastal-influenced watersheds, with peak discharge from snowmelt occurring between May and July, and additional high flows from fall precipitation and rain-on-snow events (Figure 2.4).

2.8 Zymoetz River

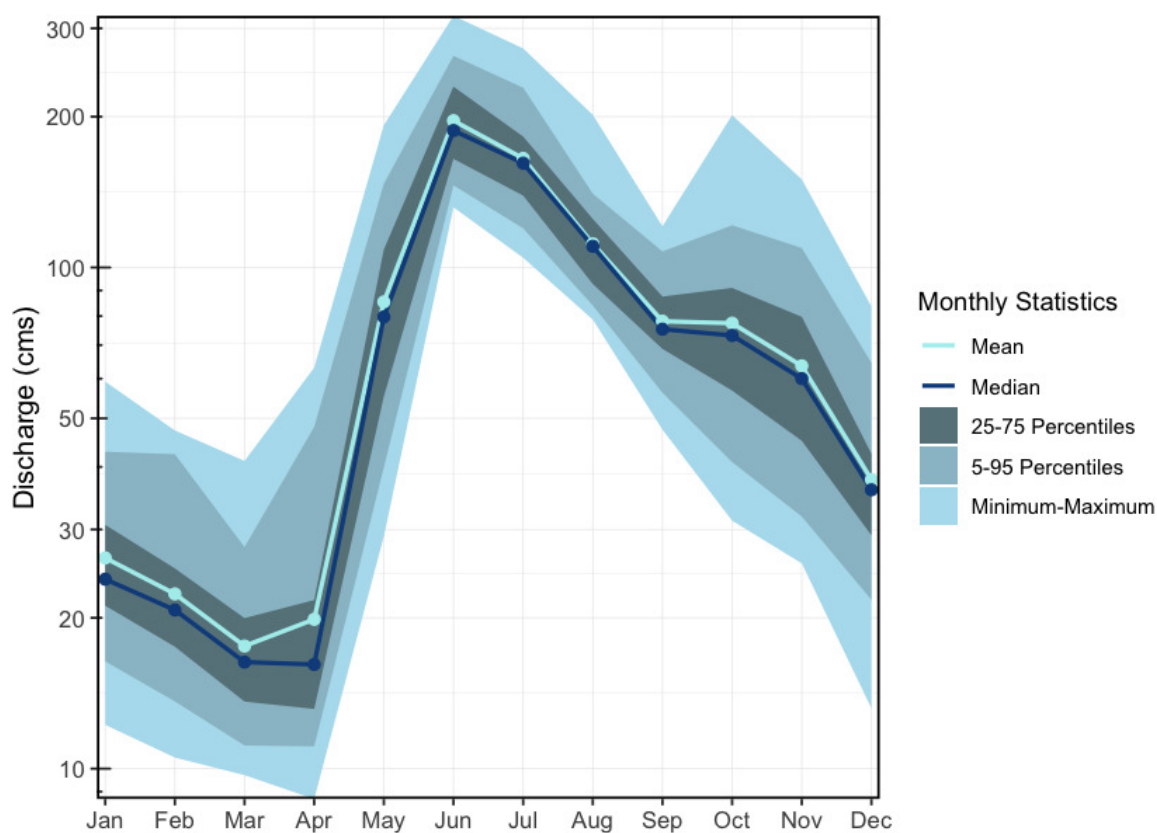


Figure 2.4: Hydrograph for the Morice River near outlet of Morice Lake (Station #08ED002). Available mean daily discharge data from 1961 to 2023.

2.8 Zymoetz River

The Zymoetz River (known locally as the Copper River) is an 8th order stream that drains an area of 3,026km² in a generally westerly direction and is a major tributary to the Skeena River, contributing approximately 10% of its flow. Flowing roughly 120km from its headwaters near Hudson Bay Mountain west of Smithers, to its confluence with the Skeena River about 8km northeast of Terrace, with headwater lakes including Aldrich, Dennis, and McDonell lakes. Elevations in the watershed range from 120m at the confluence, to 2740m in the Howson Range. The watershed is accessed via logging roads from Highway 16 near Smithers and Terrace, though access to the mid-watershed is limited due to road washouts. The Duthie Mine, active during the 1930s and 1950s on the southwest slope of Hudson Bay Mountain, has been linked to contamination in surrounding streams and lakes (Gottesfeld et al. 2002), while the lower watershed has experienced riparian habitat loss due to wildfires, forestry activity, and pipeline and road construction (Gottesfeld et al. 2002). Snowmelt dominates the hydrology, with peak flows from May to early June and a mean annual discharge of 105m³/s at station 08EF005 near Smithers (Figure 2.5).

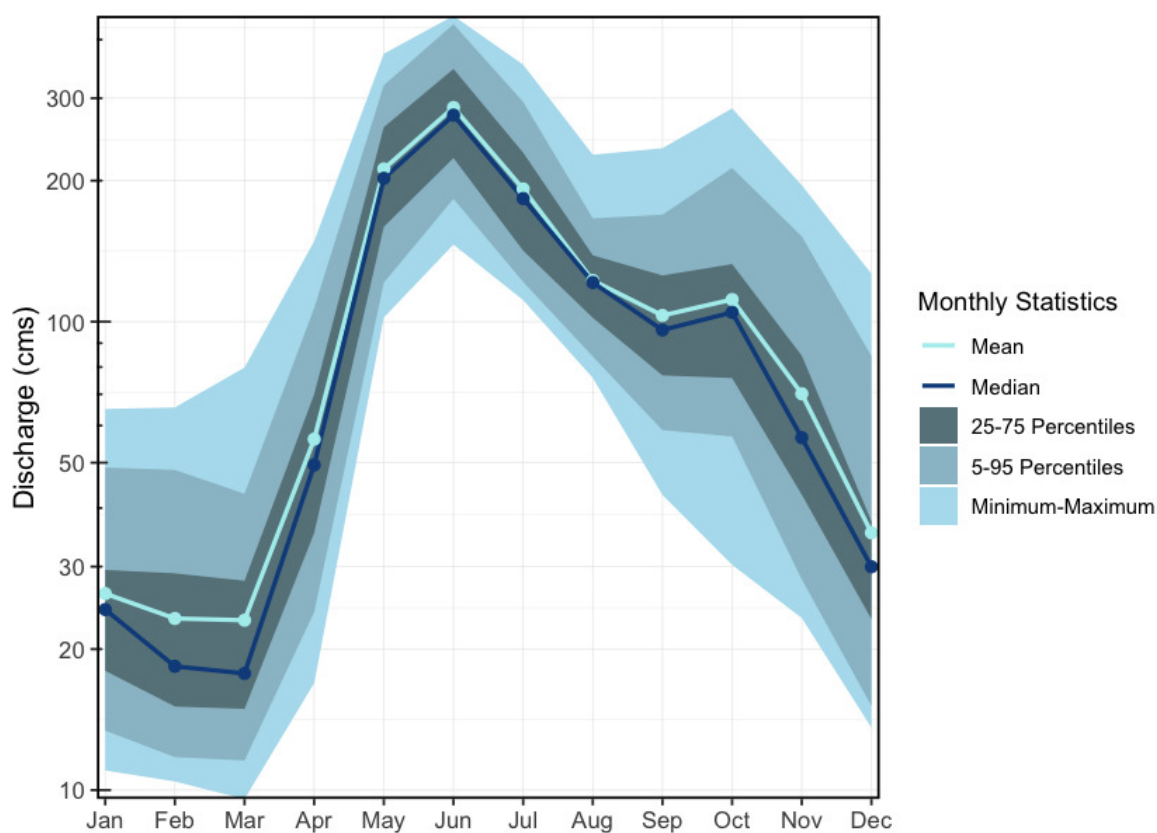


Figure 2.5: Hydrograph for the Zymoetz River above O.K. Creek (Station #08EF005 - Lat 54.49363 Lon -128.32466). Available mean daily discharge data from 1963 to 2022.

2.9 Kispiox River

The Kispiox River is a 7th order stream that flows approximately 140km southeast to its confluence with the Skeena River near Kispiox Village, draining a watershed area of 2,100km² and contributing an estimated 9% of the Skeena River's flow. Elevations in the watershed range from 200m at the mouth to 2,090m on Kispiox Mountain. The river is primarily fed by glacier and high-elevation snowmelt, with peak discharge occurring in May and June (Figure 2.6), and a mean annual discharge of 45m³/s recorded at station 08EB004 near Hazelton. Important sockeye systems in the upper watershed include Swan and Stephens lakes; Swan Lake drains via Club Lake into Stephens Lake, which flows via Stephens Creek into the Kispiox mainstem. The upper third of the watershed, upstream of the Nangeese River, is relatively undisturbed due to limited road access and protection within Swan Lake Kispiox River Provincial Park (Gottesfeld et al. 2002). Reported threats to aquatic ecosystems in the Kispiox Valley include erosion, obstructions, sedimentation, and altered water yield.

2.9 Kispiox River

The Kispiox watershed is also in danger due to the proposed Prince Rupert Gas Transmission (PRGT) pipeline which would run approximately 900km from Hudson's Hope in northeast BC to the proposed Ksi Lisims LNG facility near the Nass River estuary. The proposed pipeline would not only cut through Kispiox River, but also the Babine, Suskwa, Salmon/Shegunia, Skeena, and Nass rivers, all of which have salmon and steelhead habitat ("LNG Pipelines I Skeena Watershed Conservation Coalition," n.d.). The project has raised significant concern due to risks to water quality, fish habitat, and indigenous rights. The Kispiox Band, Skeena Watershed Conservation Coalition, Kispiox Valley Community Centre Association, and other local groups have publicly opposed the project and launched legal action challenging the BC Energy Regulator's permitting decisions. As of March 2025, the pipeline had not yet been constructed, and legal proceedings remain ongoing ("Groups Challenge Rush Start of Construction on Pipeline" 2025).

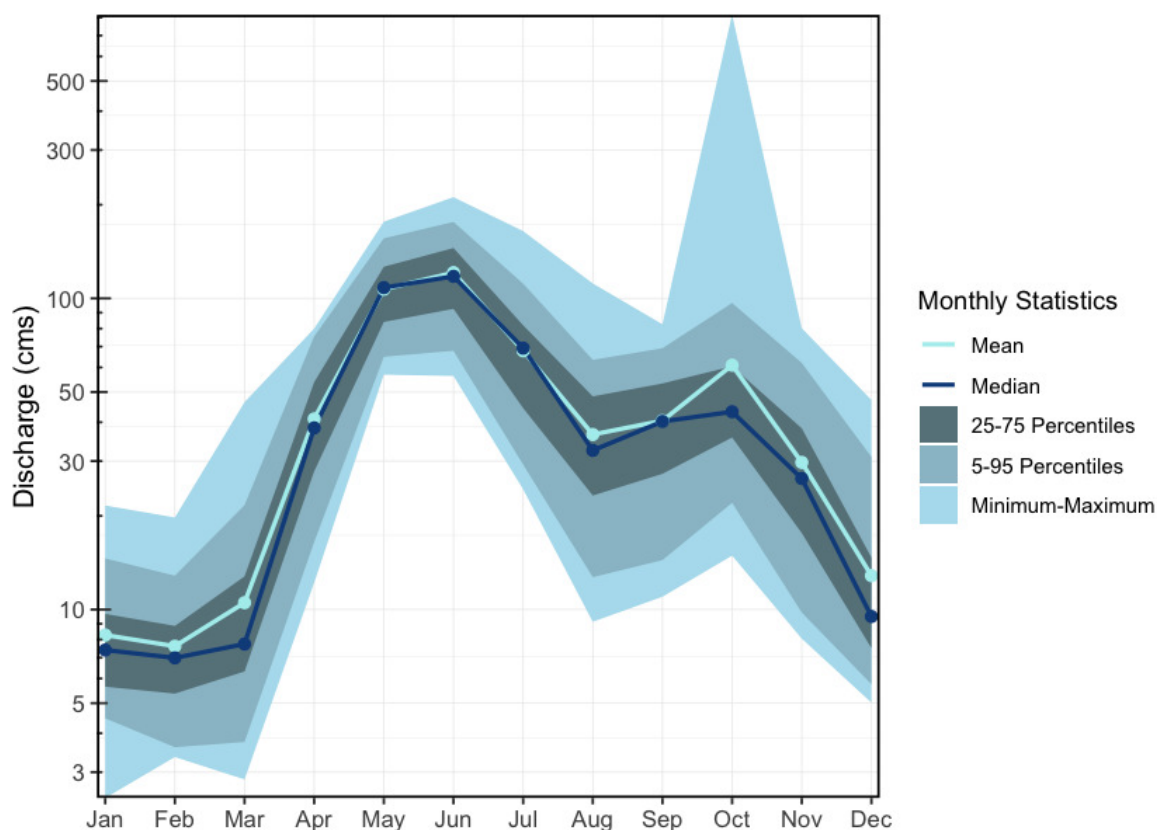


Figure 2.6: Hydrograph for the Kispiox River near Hazelton (Station #08EB004 - Lat 55.43385 Lon -127.71616). Available mean daily discharge data from 1963 to 2024.

2.10 Kitsumkalum River

The Kitsumkalum River is a 6th order stream that flows north-east from the Coast Mountains into Kitsumkalum Lake, then south to its confluence with the Skeena River at Terrace, draining a watershed area of 2,289km². Major tributaries include the Cedar River, Nelson River, Mayo Creek, Goat Creek, Lean-To Creek, and Deep Creek (McElhanney 2022). Peak flows occur in May–June from snowmelt, with additional peaks in the fall due to rainfall events (McElhanney 2022). A hydrometric station located below Kitsumkalum Lake has been active since 2018 and records a mean annual discharge of 123m³/s (Figure [2.7](#)).

The Kitsumkalum River watershed has been highly impacted by logging. Many of the tributaries to the Kitsumkalum River have altered channel morphology, increased bedload movement, bank failures, sediment loading, and debris accumulation (Gottesfeld and Rabnett 2007).

There has been a significant amount of work done to enhance salmon populations within the watershed. The SkeenaWild Conservation Trust is conducting riparian restoration surveys on several tributaries to the Kitsumkalum River, including Willow Creek, Spring Creek, Lean-To Creek, and Deep Creek (Healthy Watersheds Initiative 2021). The Deep Creek Hatchery, operated by the Terrace Salmonid Enhancement Society, has been supporting Kitsumkalum River chinook populations since 1984 (Gottesfeld and Rabnett 2007). Additionally, there is a small groundwater facility for the incubation and rearing coho and chum, run by the Kitsumkalum First Nation (Gottesfeld and Rabnett 2007). In 2000, the Clear Creek Eastern Side Channel was constructed to enhance juvenile rearing habitat and adult spawning habitat for coho salmon on Clear Creek, a tributary to the Kitsumkalum River, however the site has not been maintained and beaver activity has obstructed fish accessibility to much of the channel (Elmer 2021).

2.11 Fisheries

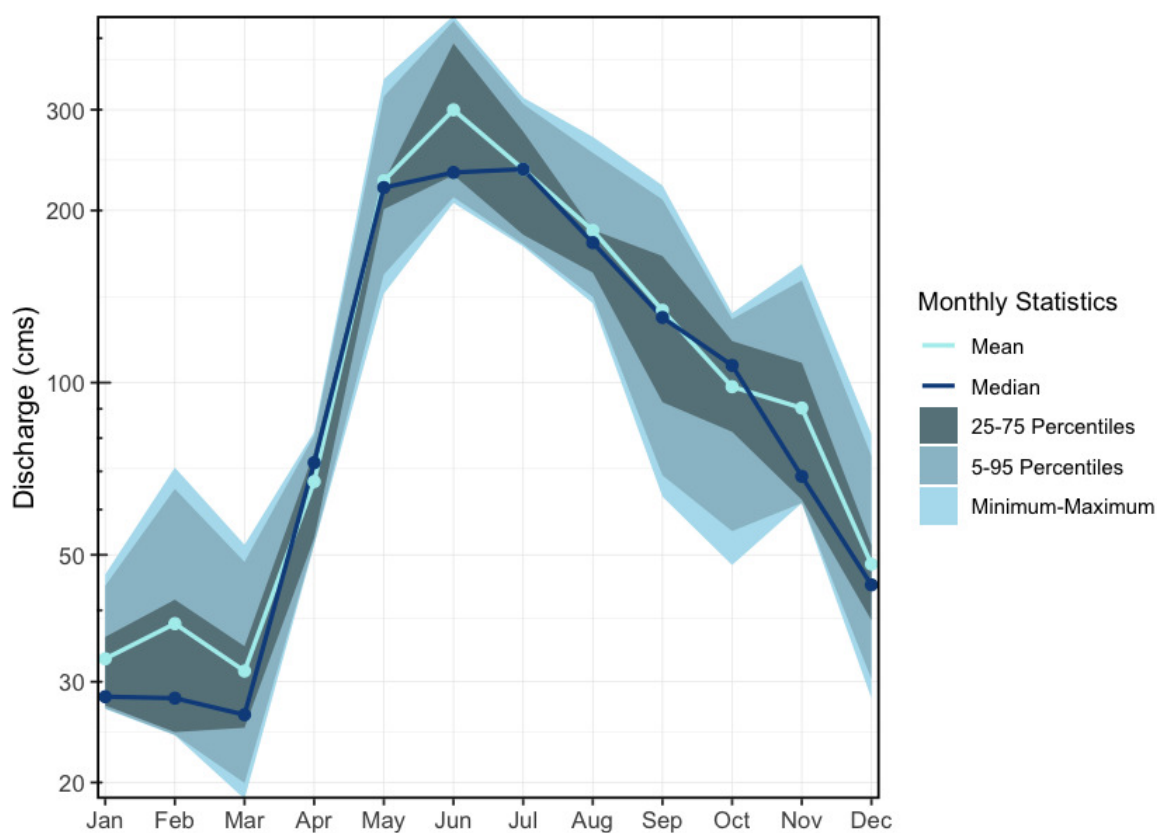


Figure 2.7: Hydrograph for the Kitsumkalum River Below Alice Creek (Station #08EG019 - Lat 54.6793 Lon -128.74396). Available mean daily discharge data from 2018 to 2023.

2.11 Fisheries

In 2004, IBM Business Consulting Services (2006) estimated the value of Skeena Fisheries at an annual average of \$110 million dollars. The Bulkley-Morice watershed is an integral part of the salmon production in the Skeena drainage and supports an internationally renown steelhead, chinook and coho sport fishery (Tamblyn 2005).

2.11.1 Bulkley River

Traditionally, the salmon stocks passing through and spawning in the greater Bulkley River were the principal food source for the Gitksan and Wet'suwet'en people living there (Wilson and Rabnett 2007). Anadromous lamprey passing through and spawning in the upper Bulkley River were traditionally also an important food source for the Wet'suwet'en (Gottesfeld and Rabnett (2007); pers comm. Mike Ridsdale, Environmental Assessment Coordinator, Office of the Wet'suwet'en). Gottesfeld and Rabnett (2007) report sourcing information from Department of Fisheries and Oceans (1991) that principal spawning areas for chinook in the Neexdzii Kwah include the mainstem above and below Buck and McQuarrie Creeks, between Cesford and Watson Creeks, and the reaches upstream and downstream of Bulkley Falls.

Renowned as a world class recreational steelhead and coho fishery, the greater Bulkley River receives some of the heaviest angling pressure in the province. In response to longstanding angler concerns with respect to overcrowding, quality of experience and conflict amongst anglers, an Angling Management Plan was drafted for the river following the initiation of the Skeena Quality Waters Strategy process in 2006 and an extensive multi-year consultation process. The plan introduces a number of regulatory measures with the intent to provide Canadian resident anglers with quality steelhead fishing opportunities. Regulatory measures introduced with the Angling Management Plan include prohibited angling for non-guided non-resident aliens on Saturdays and Sundays, Sept 1 - Oct 31 within the Bulkley River, angling prohibited for non-guided non-resident aliens on Saturdays and Sundays, all year within the Suskwa River and angling prohibited for non-guided non-resident aliens Sept 1 - Oct 31 in the Telkwa River. The Neexdzii Kwah is considered Class II water and there is no fishing permitted upstream of the Morice/Bulkley River Confluence (FLNRO 2013a, 2013b; FLNRORD 2019).

2.11.1.1 Upper Bulkley Falls

A detailed field assessment and write up regarding the upper Bulkley falls was conducted as part of fish passage restoration in the watershed - is presented in Irvine (2021) with a condensed summary here. The site was assessed on October 28, 2021 by Nallas Nikal, B.i.T, and Chad Lewis, Environmental Technician. The top of the falls is located at 11U.678269.6038266 at an elevation of 697m approximately 11.3km downstream of Bulkley Lake and upstream of Ailport Creek. Within the Bulkley River immediately below the 12 - 15m high bedrock falls, the channel width was 17.4m and the wetted width was 15.6m. Two channels comprised the falls. The primary channel was 20m long, had a channel/wetted width of 8.5m, a 16% grade and water depths ranging from 35 - 63cm. The secondary channel was 25m long, with channel/wetted widths of 7.5m, a grade of 12% and water depths ranging from 3 - 13cm (Irvine 2021).

Dyson (1949) and Stokes (1956) report substantial use of habitat above Bulkley Falls by steelhead, chinook, coho and sockeye utilization in the past (pre-1950) based on spawning reports. Both authors concluded that the Bulkley Falls pose a partial obstruction to migrating fish based on flow levels. Chinook, which migrate early in the summer when water levels are high, have been noted as able to ascend the falls in normal to high water years and in high water years it was thought that coho and steelhead could ascend. Gottesfeld and Rabnett (2007) report that the falls are almost completely impassable to all salmon during low water flows. Stokes (1956) reports that there was high value spawning habitat located within the first 3km of the Neexdzii Kwah from the outlet of Bulkley Lake.

Wilson and Rabnett (2007) reported that approximately 11.3 km downstream of the Bulkley Lake outlet and just upstream of Watson Creek, the upper Bulkley falls is an approximately 4m high narrow rock sill that crosses the Neexdzii Kwah, producing a steep cascade section. This obstacle to fish passage is recorded as an almost complete barrier to fish passage for salmon during low

2.11 Fisheries

water flows. Wilson and Rabnett (2007) also reported that coho have not been observed beyond the falls since 1972.

2.11.2 Morice River

Detailed reviews of Morice River watershed fisheries can be found in Bustard and Schell (2002), Gottesfeld et al. (2002), Schell (2003), Gottesfeld and Rabnett (2007), and ILMB (2007) with a comprehensive review of water quality by Oliver (2018). Overall, the Morice watershed contains high fisheries values as a major producer of chinook, pink, sockeye, coho and steelhead.

2.11.3 Zymoetz River

Within the Zymoetz Watershed, there are many areas with high fishery values. Steelhead are the most extensively documented fish species in the Zymoetz River watershed. Adults enter the river from July to November and then go on to spawn the following year in late spring to early summer. The Zymoetz River is a relatively steep system. Two canyons are located 6.4 and 19.6 kilometers upstream of the Skeena River confluence. These canyons make access to the Zymoetz difficult for pink and chum salmon (Gottesfeld et al. 2002). The Zymoetz River is renowned for its aggressive steelhead that have been known to take flies or lures. There is a 50km stretch upstream of Limonite Creek that's very remote and offers high quality fishing opportunities for anglers (FLNRORD 2013).

Traditional First Nations use of the upper Zymoetz River watershed by the Gitksan and Wet'suwet'en people differed between community sites, residences, and fish houses, and was large and diverse. From the upper to lower Zymoetz River and to the Skeena River, a significant ancient grease trail connected, with a branch track forking through Limonite Creek and flowing down the Telkwa River. The fishery used a weir at the mouth of McDonell Lake and spears at Six Mile Flats, near Dennis Lake. There is no information on native fisheries on the lower Zymoetz River. The Zymoetz is considered to be one of the top ten steelhead rivers in BC (Gottesfeld et al. 2002).

2.11.4 Kispiox River

Kispiox River salmon are an important food source and cultural symbol for the Gitksan people with sockeye and coho historically the two most significant species. Gitangwalk and Lax Didax, two significant villages that were both abandoned in the early 1900s, were situated on the Kispiox in such a way as to block the sockeye and coho salmon's upstream migration to the Upper Kispiox River spawning grounds providing opportunities to gather and preserve a significant amount of high-quality food over relatively short time periods (Gottesfeld et al. 2002). The 100 km of mainstem and 300km of tributary streams in the Kispiox River Watershed are considered high value fish habitat supporting migration, spawning and rearing for many fish species. The Kispiox fisheries supports both recreational and commercial fishing while also enhancing the ecology, nutrient regime, and structural diversity of the drainage. Since 1992, sockeye and coho escapements from the Kispiox Watershed have been documented by the Gitksan Watershed Authorities as they create strong cultural, economic, and symbolic ties for the local communities (Gottesfeld et al. 2002).

2.11.5 Kitsumkalum River

The Kitsumkalum River is an important waterway for all species of salmon. It is one of the three main chinook producing rivers in the Skeena watershed and supports all five species of pacific salmon, steelhead, and other resident trout and char species (McElhanney 2022). Most notably, the Kitsumkalum River has consistently produced the largest-bodied chinook in the Skeena Watershed, as well as on most of the Pacific coast (Gottesfeld and Rabnett 2007). The watershed supports strong recreational coho, steelhead, and chinook fishing. Kitsumkalum salmon also play an important role in the culture and economy of the Kitsumkalum Band (Gottesfeld and Rabnett 2007).

2.11.6 Salmon Stock Assessment Data

Fisheries and Oceans Canada stock assessment data for the study area was accessed via the [NuSEDS-New Salmon Escapement Database System](#) through the [Open Government Portal](#) (Fisheries and Oceans Canada, n.d.). Records are presented in [Appendix - Salmon Stock Assessment Data \(page 45\)](#).

2.11.7 Fish Species

Fish species recorded in the study area watershed groups are detailed in [Appendix - Fish Species \(page 47\)](#) (MoE 2024a). Coastal cutthroat trout and bull trout are considered of special concern (blue-listed) provincially. Summaries of some of the Skeena fish species life history, biology, stock status, and traditional use are documented in Schell (2003), Wilson and Rabnett (2007), Gottesfeld et al. (2002) and Office of the Wet'suwet'en (2013). Wilson and Rabnett (2007) discuss chinook, pink, sockeye, coho, steelhead and indigenous freshwater Bulkley River fish stocks within the context of key lower and upper Bulkley River habitats such as the Suskwa River, Station Creek, Harold Price Creek, Telkwa River and Buck Creek. Key areas within the upper Bulkley River watershed with high fishery values, documented in Schell (2003), are the upper Bulkley mainstem, Buck Creek, Dungate Creek, Barren Creek, McQuarrie Creek, Byman Creek, Richfield Creek, Johnny David Creek, Aitken Creek and Emerson Creek.

Some key areas of high fisheries values for chinook, sockeye and coho are noted in Bustard and Schell (2002) as McBride Lake, Nanika Lake, and Morice Lake watersheds. A draft gantt chart for select species in the Morice River watershed were derived from reviews of the aforementioned references and is included as Figure [2.8](#). The data is considered in draft form and will be refined over the spring and summer of 2021 with local fisheries technicians and knowledge holders during the collaborative assessment planning and fieldwork activities planned.

In the 1990's the Morice River watershed, Gottesfeld and Rabnett (2007) estimated that chinook comprised 30% of the total Skeena system chinook escapements. It is estimated that Morice River coho comprise approximately 4% of the Skeena escapement with a declining trend noted since the 1950 in Gottesfeld and Rabnett (2007). Coho spawn in major tributaries and small streams ideally

2.11 Fisheries

at locations where downstream dispersal can result in seeding of prime off channel habitats including warm productive sloughs and side channels. Of all the salmon species, coho rely on small tributaries the most (Bustard and Schell 2002). Bustard and Schell (2002) report that much of the distribution of coho into non-natal tributaries occurs during high flow periods of May - early July with road culverts blocking migration into these habitats.

Summaries of historical fish observations in the Bulkley River and Morice River watershed groups (n=4033), graphed by remotely sensed average gradient as well as measured or modelled channel width categories for their associated stream segments where calculated with `bcfishpass` and `bcfishobs` and are provided in Figures [2.9](#) - [2.10](#).

Fish species recorded in the study area watershed groups are presented in [Appendix - Fish Species \(page 47\)](#).

2 Background

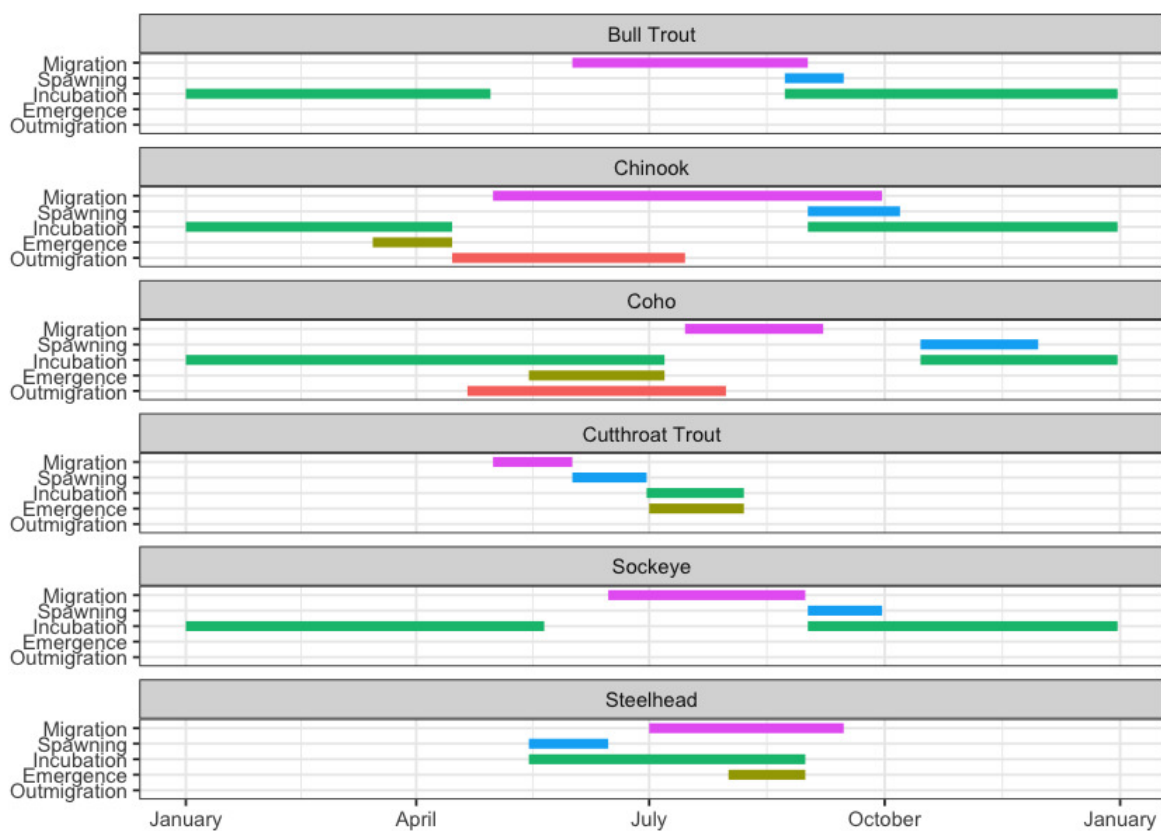


Figure 2.8: Gantt chart for select species in the Morice River watershed. To be updated in consultation with local fisheries technicians and knowledge holders.

2.11 Fisheries

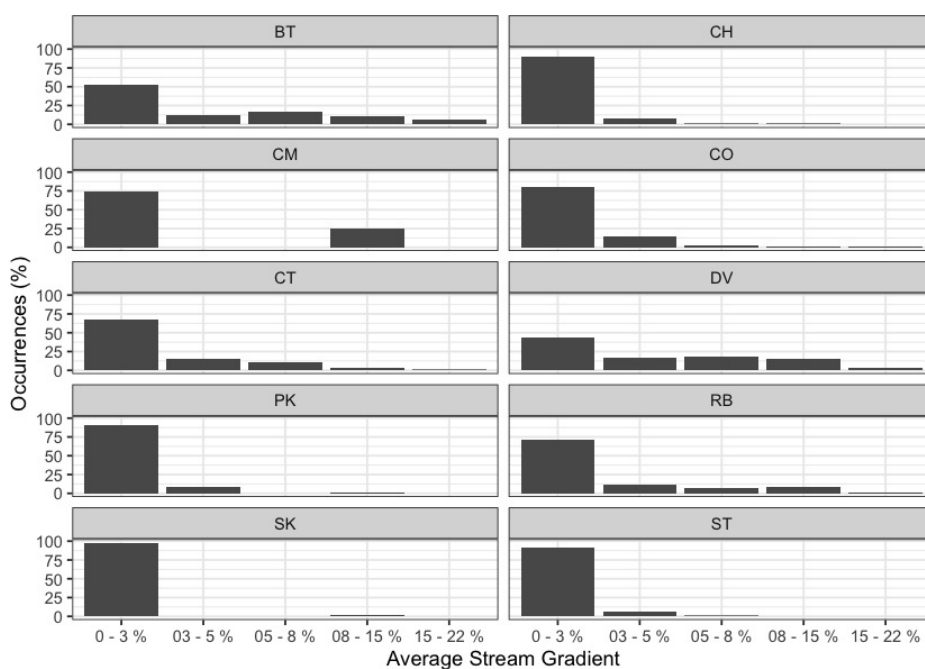


Figure 2.9: Summary of historic salmonid observations vs. stream gradient category for the Bulkley River watershed group.

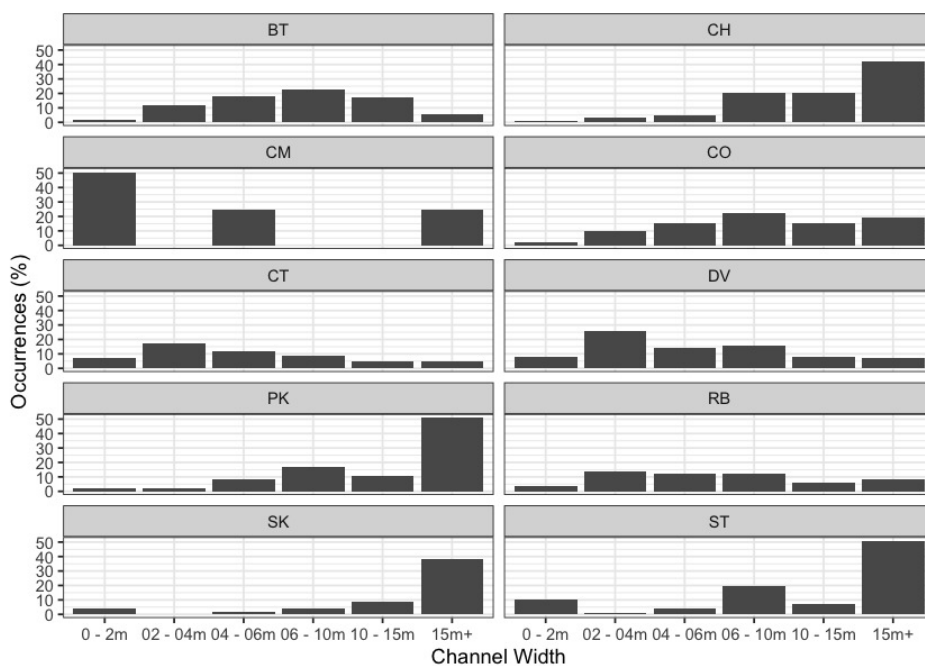


Figure 2.10: Summary of historic salmonid observations vs. channel width category for the Bulkley River watershed group.

3 Methods

3.1 Collaborative GIS Environment

Geographical Information Systems are essential for developing and communicating restoration plans as well as the reasons they are required and how they are developed. Without the ability to visualize the landscape and the data that is used to make decisions, it is difficult to conduct and communicate the need for restoration, the details of past and future plans as well as and the potential results of physical works.

To facilitate the planning and implementation of restoration activities, a collaborative GIS environment has been established using [QGIS](#) and is served on the cloud using source code stored [here](#). This environment is intended to be a space where project team members can access, view, and contribute to the amalgamation of background spatial data and the development of restoration as well as monitoring for the project. The collaborative GIS environment allows users to view, edit, and analyze shared, up to date spatial data on personal computers in an office setting as well as on phones and tablets in the field. At the time of reporting, the environment was being used to develop and share maps, conduct spatial analyses, communicate restoration plans to stakeholders as well as to provide a central place to store methodologies and tools for conducting field assessments on standardized pre-developed digital forms. The platform can also be used to track the progress of restoration activities and monitor changes in the landscape over time, helping encourage the record keeping of past and future restoration activities in a coordinated manner.

The shared QGIS project was created using scripts currently kept in [dff-2022](#) with the precise calls to project creation scripts tracked in the `project_creation_and_permissions.txt` document kept in the main QGIS project directory. Information about the scripts used for GIS project creation and updates can be viewed [here](#) with outcomes of their use summarized below:

- Download and clip user specified layers from the [BC Data Catalogue](#) as well as data layers stored in custom Amazon Web Services buckets for an area of interest defined by a list of watershed groups and load to a geopackage called `background_layers.gpkg` stored in the main directory of the project.
- A project directory is created to hold the spatial data and QGIS project information (ie. layer symbology and naming conventions, metadata, etc.).
- Metadata for individual project spatial layers is kept in the `rfp_tracking` table within the `background_layers.gpkg` along with tables related to user supplied stream width/gradient inputs to `bcfishpass` to model potentially high value habitat that is accessible to fish species of interest.

3.1.1 Mapping

The workflows to produce the georeferenced pdf maps include using a QGIS layer file defining and symbolizing all layers required and are continuously evolving. At the time of reporting - mapping scripts and associated layer file were kept under version control within `bcfishpass` [here](#). Loading

the QGIS layer file within a QGIS project, allows load and representation of all map component layers provided the user points to a postgresql database populated via `bcfishpass` outputs.

3.2 Planning — Habitat and Connectivity Modelling

3.2.1 Habitat Modelling

Habitat modelling used to help guide planning for field assessments is generated by `bcfishpass` (Norris [2020] 2024) which has been designed to prioritize potential fish passage barriers for assessment or remediation by generating a simple model of aquatic habitat connectivity. We utilize the `bcfishpass` access model, linear spawning/rearing habitat model and lateral habitat connectivity for planning purposes. These models provide a valuable starting point, but their results are not definitive and should always be considered with professional judgment. Detailed information regarding model methodology, select parameters and known model limitations are detailed in Norris ([2020] 2024) with key documentation linked below:

- [Access model](#)
- [Linear spawning/rearing habitat models](#)
- [Lateral habitat model](#)

Table 3.1 documents the custom species-specific thresholds for stream gradient and channel width applied to the linear spawning and rearing habitat model for this year’s project planning.

Table 3.1: Stream gradient and channel width thresholds used to model potentially highest value fish habitat.

Variable	Chinook Salmon	Chum Salmon	Coho Salmon	Pink Salmon	Sockeye Salmon	Steelhead
Spawning Gradient Max (%)	4.5	6.5	5.5	6.5	2.5	4.5
Spawning Width Min (m)	4.0	2.1	2.0	2.1	2.0	4.0
Rearing Width Min (m)	1.5	–	1.5	–	1.5	1.5
Rearing Gradient Max (%)	5.5	–	5.5	–	–	8.5
*						

3.2.1.1 Statistical Support for `bcfishpass` Fish Habitat Modelling Updates

This project provided the statistical background for updates to `bcfishpass` that facilitated incorporation of channel width (observed or predicted) into species specific linear spawning/rearing habitat models. In early 2021, Bayesian statistical methods were developed to predict channel width in all provincial freshwater atlas stream segments where width measurements had not previously been measured in the field. The model was based on the relationship between watershed area and mean annual precipitation weighted by upstream watershed area (Thorley and Irvine 2021). In December of 2021, Thorley and Irvine (2021) methods were updated using a power model derived by Finnegan et al. (2005) which relates stream discharge to watershed area and mean annual precipitation resulting in Thorley et al. (2021) which was utilized for channel width estimates within

bcfishpass modelling at the time of reporting. More detailed documentation of the methodology used to facilitate both the data collection and statistical analysis can be sourced in Irvine ([2021] 2022) and Thorley et al. (2021).

In 2024, in collaboration with Poisson Consulting - stream discharge and temperature causal effects pathways were mapped with the intent of focusing aquatic restoration actions in areas of highest potential for positive impacts on fisheries values (ie. elimination of areas from intrinsic models where water temperatures are likely too cold to support fish production). The project began with a custom mechanistic model (visually represented [here](#)), but the model struggled to converge. The project then shifted to the air2stream model, which offers a middle ground between fully mechanistic models—often data-intensive and reliant on quantities that are difficult to measure or estimate—and purely statistical models, which lack physical justification and perform poorly when extrapolated to new conditions (Toffolon and Piccolroaz (2015)). After several adaptations, the expected stream temperatures were best modeled using the four-parameter version of the air2stream model, with added random effects by site for each of the four parameters (Hill et al. (2024)). The data used for the model were sourced from the following locations, for years 2019-2021:

- Water temperature data collected in the Nechako Watershed were downloaded from [Zenodo](#) (Gilbert et al. 2022).
- Hourly air temperature data were obtained from the ERA-5-Land dataset via the Copernicus Climate Change Service (Muñoz Sabater (2019))
- Daily baseflow and surface runoff data were sourced from the [Pacific Climate Impacts Consortium's Gridded Hydrologic Model Output](#) using the ACCESS1-0_rcp85 scenario (Pacific Climate Impacts Consortium (n.d.)).

3.3 Fish Passage Assessments

3.3.1 Natural Barriers to Fish Passage

Our assessments may include natural features such as waterfalls that could limit fish passage. This informs whether upstream culvert upgrades would restore access for anadromous species (e.g., salmon) or primarily benefit resident fish already upstream. We document these features by measuring height, gradient, and pool depth, recording field observations, capturing site photographs, and reviewing background sources for context.

3.3.2 Road Stream Crossings

In the field, crossings prioritized for follow-up were first assessed for fish passage following the procedures outlined in “Field Assessment for Determining Fish Passage Status of Closed Bottomed Structures” (MoE 2011). The reader is referred to (MoE 2011) for detailed methodology. Crossings surveyed included closed bottom structures (CBS), open bottom structures (OBS) and crossings considered “other” (i.e. fords). Photos were taken at surveyed crossings and when possible included images of the road, crossing inlet, crossing outlet, crossing barrel, channel downstream and channel upstream of the crossing and any other relevant features. The following information was recorded for all surveyed crossings: date of inspection, crossing reference, crew member initials, Universal Transverse Mercator (UTM) coordinates, stream name, road name and kilometer, road tenure information, crossing type, crossing subtype, culvert diameter or span for OBS, culvert length or width for CBS. A more detailed “full assessment” was completed for all closed bottom structures and included the following parameters: presence/absence of continuous culvert embedment (yes/no), average depth of embedment, whether or not the culvert bed resembled the native stream bed, presence of and percentage backwatering, road fill depth, outlet drop, outlet pool depth, inlet drop, culvert slope, average downstream channel width, stream slope, presence/absence of beaver activity, presence/absence of fish at time of survey, type of valley fill, and a habitat value rating. Habitat value ratings were based on channel morphology, flow characteristics (perennial, intermittent, ephemeral), fish migration patterns, the presence/absence of deep pools, un-embedded boulders, substrate, woody debris, undercut banks, aquatic vegetation and overhanging riparian vegetation (Table 3.2).

Table 3.2: Habitat value criteria (Fish Passage Technical Working Group, 2011).

Habitat Value	Fish Habitat Criteria
High	The presence of high value spawning or rearing habitat (e.g., locations with abundance of suitably sized gravels, deep pools, undercut banks, or stable debris) which are critical to the fish population.
Medium	Important migration corridor. Presence of suitable spawning habitat. Habitat with moderate rearing potential for the fish species present.
Low	No suitable spawning habitat, and habitat with low rearing potential (e.g., locations without deep pools, undercut banks, or stable debris, and with little or no suitably sized spawning gravels for the fish species present).

3.3 Fish Passage Assessments

Fish passage potential was determined for each stream crossing identified as a closed bottom structure as per MoE (2011). The combined scores from five criteria: depth and degree to which the structure is embedded, outlet drop, stream width ratio, culvert slope, and culvert length were used to screen whether each culvert was a likely barrier to some fish species and life stages (Table 3.3, Table 3.4). These criteria were developed based on data obtained from various studies and reflect an estimation for the passage of a juvenile salmon or small resident rainbow trout (Clarkin et al. 2005; Bell 1991; Thompson 2013). For crossings determined to be potential barriers or barriers based on the data, a culvert fix and recommended diameter/span was proposed.

Table 3.3: Fish Barrier Risk Assessment (MoE 2011).

Risk	LOW	MOD	HIGH
Embedded	>30cm or >20% of diameter and continuous	<30cm or 20% of diameter but continuous	No embedment or discontinuous
Value	0	5	10
Outlet Drop (cm)	<15	15-30	>30
Value	0	5	10
SWR	<1.0	1.0-1.3	>1.3
Value	0	3	6
Slope (%)	<1	1-3	>3
Value	0	5	10
Length (m)	<15	15-30	>30
Value	0	3	6

Table 3.4: Fish Barrier Scoring Results (MoE 2011).

Cumulative Score	Result
0-14	passable
15-19	potential barrier
>20	barrier

The habitat gain index is the quantity of modelled habitat upstream of the subject crossing and represents an estimate of habitat gained with remediation of fish passage at the crossing. For this project, a gradient threshold between accessible and non-accessible habitat was set at 20% (for a minimum length of 100m) intended to represent the maximum gradient of which the strongest swimmers of anadromous species (steelhead) are likely to be able to migrate upstream.

3 Methods

For reporting of Phase 1 - fish passage assessments within the body of this report (Table 3.3), a “total” value of habitat <20% output from *bcfishpass* was used to estimate the amount of habitat upstream of each crossing less than 20% gradient before a falls of height >5m - as recorded in MoE (2024b) or documented in other *bcfishpass* online documentation. For Phase 2 - habitat confirmation sites, conservative estimates of the linear quantity of habitat to be potentially gained by fish passage restoration, steelhead rearing maximum gradient threshold (8.5%) was used. To generate estimates for area of habitat upstream (m^2), the estimated linear length was multiplied by half the downstream channel width measured (overall triangular channel shape) as part of the fish passage assessment protocol. Although these estimates are not generally conservative, have low accuracy and do not account for upstream stream crossing structures they allow a rough idea of the best candidates for follow up.

Potential options to remediate fish passage were selected from MoE (2011) and included:

- Removal (RM) - Complete removal of the structure and deactivation of the road.
- Open Bottom Structure (OBS) - Replacement of the culvert with a bridge or other open bottom structure. Based on consultation with FLNR road crossing engineering experts, for this project we considered bridges as the only viable option for OBS type .
- Streambed Simulation (SS) - Replacement of the structure with a streambed simulation design culvert. Often achieved by embedding the culvert by 40% or more. Based on consultation with FLNR engineering experts, we considered crossings on streams with a channel width of <2m and a stream gradient of <8% as candidates for replacement with streambed simulations.
- Additional Substrate Material (EM) - Add additional substrate to the culvert and/or downstream weir to embed culvert and reduce overall velocity/turbulence. This option was considered only when outlet drop = 0, culvert slope <1.0% and stream width ratio < 1.0.
- Backwater (BW) - Backwatering of the structure to reduce velocity and turbulence. This option was considered only when outlet drop < 0.3m, culvert slope <2.0%, stream width ratio < 1.2 and stream profiling indicates it would be effective..

3.3.3 Cost Estimates

Cost estimates for structure replacement with bridges and embedded culverts were generated based on the channel width, slope of the culvert, depth of fill, road class and road surface type. Road details were sourced from FLNRORD (2020b) and FLNRORD (2020a) through *bcfishpass*. Interviews with Phil MacDonald, Engineering Specialist FLNR - Kootenay, Steve Page, Area Engineer - FLNR - Northern Engineering Group and Matt Hawkins - MoTi - Design Supervisor for Highway Design and Survey - Nelson were utilized to help refine estimates which have since been adjusted for inflation in 2020 and based on past experience.

Base costs for installation of bridges on forest service roads and permit roads with surfaces specified in provincial GIS road layers as rough and loose was estimated at \$30000/linear m and

3.3 Fish Passage Assessments

assumed that the road could be closed during construction and a minimum bridge span of 15m. For streams with channel widths <2m, embedded culverts were reported as an effective solution with total installation costs estimated at \$100k/crossing (pers. comm. Phil MacDonald, Steve Page then adjusted for inflation in 2020). For larger streams (>6m), estimated span width increased proportionally to the size of the stream. For crossings with large amounts of fill (>3m), the replacement bridge span was increased by an additional 3m for each 1m of fill >3m to account for cutslopes to the stream at a 1.5:1 ratio. To account for road type, a multiplier table was generated to estimate incremental cost increases with costs estimated for structure replacement on paved surfaces, railways and arterial/highways costing up to 15 times more than forest service roads due to expenses associate with design/engineering requirements, traffic control and paving. The cost multiplier table (Table 3.5) should be considered very approximate with refinement recommended for future projects.

Table 3.5: Cost multiplier table based on road class and surface type.

Class	Surface	Class Multiplier	Surface Multiplier	Bridge \$/15m	Streambed Simulation \$
FSR	Rough	1	1	450,000	100,000
FSR	Loose	1	1	450,000	100,000
Resource	Loose	1	1	450,000	100,000
Collector	Loose	1	1	450,000	100,000
Recreation	Loose	1	1	450,000	100,000
Resource	Rough	1	1	450,000	100,000
Permit	Unknown	1	1	450,000	100,000
Permit	Loose	1	1	450,000	100,000
Permit	Rough	1	1	450,000	100,000
Unclassified	Loose	1	1	450,000	100,000
Unclassified	Rough	1	1	450,000	100,000
Unclassified	Paved	1	2	750,000	150,000
Unclassified	Unknown	1	2	750,000	150,000
Local	Loose	4	1	1,500,000	200,000
Local	Paved	4	2	3,000,000	400,000
Collector	Paved	4	2	3,000,000	400,000
Arterial	Paved	15	2	11,250,000	1,500,000
Highway	Paved	15	2	11,250,000	1,500,000
Rail	Rail	15	2	11,250,000	1,500,000

3.3.4 Climate Change Risk Assessment

In collaboration with the Ministry of Transportation and Infrastructure (MoTi), a new climate change replacement program aims to prioritize vulnerable culverts for replacement (pers. comm Sean Wong, 2022) based on data collected and ranked related to three categories - culvert condition, vulnerability and priority. Within the “condition” risk category - data was collected and crossings were ranked based on erosion, embankment and blockage issues. The “climate” risk category included ranked assessments of the likelihood of both a flood event affecting the culvert as well as the consequence of a flood event affecting the culvert. Within the “priority” category the following factors were ranked - traffic volume, community access, cost, constructability, fish bearing status and environmental impacts (Table 3.6). This project is still in its early stages with methodology changes going forward.

Table 3.6: Climate change data collected at MoTi culvert sites

Parameter	Description
erosion_issues	Erosion (scale 1 low - 5 high)
embankment_fill_issues	Embankment fill issues 1 (low) 2 (medium) 3 (high)
blockage_issues	Blockage Issues 1 (0-30%) 2 (>30-75%) 3 (>75%)
condition_rank	Condition Rank = embankment + blockage + erosion
condition_notes	Describe details and rational for condition rankings
likelihood_flood_event_affecting_culvert	Likelihood Flood Event Affecting Culvert (scale 1 low - 5 high)
consequence_flood_event_affecting_culvert	Consequence Flood Event Affecting Culvert (scale 1 low - 5 high)
climate_change_flood_risk	Climate Change Flood Risk (likelihood x consequence) 1-6 (low) 6-12 (medium) 10-25 (high)
vulnerability_rank	Vulnerability Rank = Condition Rank + Climate Rank
climate_notes	Describe details and rational for climate risk rankings
traffic_volume	Traffic Volume 1 (low) 5 (medium) 10 (high)
community_access	Community Access - Scale - 1 (high - multiple road access) 5 (medium - some road access) 10 (low - one road access)
cost	Cost (scale: 1 high - 10 low)
constructability	Constructability (scale: 1 difficult -10 easy)
fish_bearing	Fish Bearing 10 (Yes) 0 (No) - see maps for fish points
environmental_impacts	Environmental Impacts (scale: 1 high -10 low)
priority_rank	Priority Rank = traffic volume + community access + cost + constructability + fish bearing + environmental impacts
overall_rank	Overall Rank = Vulnerability Rank + Priority Rank
priority_notes	Describe details and rational for priority rankings

3.4 Habitat Confirmation Assessments

Following fish passage assessments, habitat confirmations were completed in accordance with procedures outlined in the document “A Checklist for Fish Habitat Confirmation Prior to the Rehabilitation of a Stream Crossing” (Fish Passage Technical Working Group 2011). The main objective of the field surveys was to document upstream habitat quantity and quality and to determine if any other obstructions exist above or below the crossing. Habitat value was assessed based on channel morphology, flow characteristics (perennial, intermittent, ephemeral), the presence/absence of deep pools, un-embedded boulders, substrate, woody debris, undercut banks, aquatic vegetation and overhanging riparian vegetation. Criteria used to rank habitat value was based on guidelines in Fish Passage Technical Working Group (2011) (Table [3.2](#)).

During habitat confirmations, to standardize data collected and facilitate submission of the data to provincial databases, information was collected on digital field forms adapted from provincial [“Site Cards”](#). Habitat characteristics recorded included channel widths, wetted widths, residual pool depths, gradients, bankfull depths, stage, temperature, conductivity, pH, cover by type, substrate and channel morphology (among others). When possible, the crew surveyed downstream of the crossing to a minimum distance 300m and upstream to a minimum distance of 500 - 600m. Any potential obstacles to fish passage were inventoried with photos, physical descriptions and locations recorded on site cards. Surveyed routes were recorded with time-signatures on handheld GPS units.

3.4.1 Fish Sampling

3.4.1.1 Electrofishing

Fish sampling was conducted on a subset of sites when biological data was considered to add significant value to the physical habitat assessment information. Electrofishing was utilized for fish sampling according to stream inventory standards and procedures found in the Reconnaissance (1:20 000) Fish and Fish Habitat Inventory Manual (Resource Inventory Standards Committee 2001). A Haltech 2000 backpack electrofisher was used within discrete site units both upstream and downstream of the subject crossing with electrofisher settings and seconds, water quality parameters (i.e. conductivity, temperature and pH), start and end locations, length of site and wetted widths (average of a minimum of three) recorded.

3.4.1.2 Fish Handling and Processing

Captured fish were held in buckets with sufficient water to minimize stress until processing, and multiple buckets were used when catch numbers were high. For each fish captured, fork length, weight and species was recorded with results documented in the fish data submission spreadsheet.

3.4.1.3 Pit Tagging

Fish with a fork length greater than 60 mm and belonging to species approved under the scientific fish collection permit SM25-984576 were tagged with Passive Integrated Transponders (PIT tags) using the [Abdominal Cavity](#) method outlined by Biomark. To anesthetize fish prior to pit tagging, we used a solution of approximately 0.1 mL of clove oil per 1 L of water (1:10,000). This concentration was selected for its efficiency in providing effective sedation with minimal residual effects, making it ideal for studies in which fish are released back into their natural habitats (Fernandes et al. 2017). The clove oil solution was prepared in advance by dissolving pure clove oil in ethyl alcohol in a 1:9 ratio (clove oil: ethyl alcohol) to enhance solubility, then mixed into the water bucket (Fernandes et al. 2017). Fish were immersed in this solution until they reached an appropriate level of anesthesia for handling and then were tagged. To maintain needle sharpness and minimize injury risk, needles were replaced approximately every 10 fish. Each tagged fish was scanned with the PIT reader, and both the PIT tag ID and row ID were recorded. Once tagged, fish were placed into a bucket of fresh water and allowed to recover before being released back into the stream. Fish information and habitat data will be submitted to the province under scientific fish collection permit SM25-984576.

3.4.2 Environmental DNA (eDNA) sampling

3.4.2.1 eDNA Sampling and Filtration

Environmental DNA (eDNA) samples were collected using two approaches: in-stream filtration with a Haltech EMOS eDNA sampler and a grab-and-go bottle collection method following Hobbs et al. (2025). For both approaches, all water collected at a site (typically 1–3L) was treated as a single sample and filtered onto one MCE filter, resulting in one eDNA replicate per site. Typically, one sample was collected upstream and one downstream of the road–stream crossing, although in some cases only a downstream sample was collected. Field blanks were collected at the start of each sampling day using distilled water.

In-stream filtration was completed using Haltech reusable aluminum housings pre-loaded with mixed cellulose ester (MCE) filters (0.45 μ m, 1 μ m, or 1.5 μ m pore sizes). When turbidity was high, a Haltech pre-filter equipped with a 100 μ m nitex screen was installed upstream of the MCE filter to remove coarse material and maintain filtration efficiency. After filtration, aluminum housings were returned to their labelled bags and stored in a cooler with ice until the end of the day. Each evening, filters were removed from housings in a clean environment, transferred into coin envelopes placed inside labelled ziplock bags with silica desiccant, and stored in a freezer until shipment. Aluminum housings were sanitized with a bleach solution, reloaded with new filters, and stored in clean ziplock bags for future use.

At sites where the grab-and-go method was used, stream water was collected in sanitized 1L HDPE Nalgene bottles. For detailed procedures, see Hobbs et al. (2025). Bottles were stored in a cooler with ice until the end of the day and then filtered through MCE filters identical to those used for the EMOS sampler. Post-filtration handling followed the same procedures as the in-stream method, with

3.5 Engineering Design

filters transferred to labelled bags with silica desiccant and frozen until shipment. Nalgene bottles were sanitized with a bleach solution after use.

All eDNA samples were shipped frozen to the University of Northern British Columbia (UNBC) laboratory for analysis.

3.4.2.2 eDNA Extraction

eDNA was extracted in the dedicated eDNA lab at UNBC and analysed with the protocols developed by Murray and Booth (2023).

3.4.3 Open Source Reporting Framework

This report is produced using open-source tools (R, bookdown) with full version control via git, enabling iterative updates. All code, data, and revision history are publicly accessible in the [project repository](#), with ongoing tasks tracked via GitHub [issues](#) and version history maintained in the [project changelog](#).

3.4.4 Aerial Imagery

Scripted processing and serving of UAV imagery collected during the project is available at https://github.com/NewGraphEnvironment/stac_uav_bc/ (Irvine [2025] 2025). [OpenDroneMap](#) was utilized to produce orthomosaics, digital surface models (DSMs), and digital terrain models (DTMs) (OpenDroneMap Authors [2014] 2025). To support efficient web-based access - imagery products were converted to cloud-optimized GeoTIFFs (COGs) using `rio-cogeo`, then collated according to the [SpatioTemporal Asset Catalog \(STAC\)](#) specification with `pystac` and uploaded to S3 storage Amazon Web Services (2025). A `titiler` tile server was set up to facilitate interactive viewing of the orthoimagery and an Application Program Interface (API) leveraging `stac-fastapi-pgstac` is served at <https://images.a11s.one> to enable linking of collection images through QGIS as well as remote spatial and temporal querying using open source software such as `rstac` (Development Seed [2019] 2025; `stac-utils` 2025; Simoes et al. 2021).

3.5 Engineering Design

Engineering designs were signed and sealed by professional engineers. When possible - completed designs are loaded to the PSCIS data portal.

3.6 Remediations

Structure replacement was conducted by project specific contractors. If not already completed, as-built drawings will be loaded to the PSCIS data portal.

3.7 Monitoring

Monitoring of fish passage restoration sites — both proposed and completed - is essential to ensure restoration investments lead to meaningful ecological outcomes. Monitoring enables evaluation of

3 Methods

whether remediation actions improve connectivity for fish and provides critical feedback to refine future prioritization and restoration strategies.

Baseline data collection, including fish sampling and aerial surveys via drone, are core components of this monitoring. While detailed methods for these activities are included in the previous habitat confirmation section, they are also fundamental to effectiveness monitoring, as they provide context to not only facilitate prioritization and communications but also for detecting change following restoration.

To support consistent and targeted assessment, a custom field form was developed for routine effectiveness monitoring based loosely on Forest Investment Account (2003) but tailored specifically for fish passage projects. Table 3.7) outlines the monitoring metrics used.

Table 3.7: Description of monitoring metrics used for effectiveness monitoring.

Parameter	Description
Dewatering	Have the remediation works led to dewatering of the channel due to substrate aggradation or other factors?
Velocity	Are flow velocities similar to those within the natural channel? Are they expected to exceed swim speeds of particular fish species/life stages of interest?
Constriction	Have the remediation works led to constriction of the channel. Compare channel width underneath structure and within construction footprint to average channel widths upstream and downstream?
Substrate	Is the substrate within/under and adjacent to the remediated structure generally equivalent to that found upstream and downstream where natural channel conditions exist?
Riparian	What is the condition of the riparian area within the construction footprint?
UAV Flight	Was a flight conducted with unmanned aerial vehicle to document conditions at time of monitoring?
Flow_depth	What are the flow depths at the time of assessment within project footprint. Are depths expected to be sufficient to facilitate upstream passage for specific species/life stages of interest?
Stability	Does the structure appear to be stable or is there evidence of erosion/shifting?
Revegetation	How were riparian areas rehabilitated and are they improving fish habitat value?
Cover	Is cover available for fish within the construction footprint in the form of overhanging vegetation, large/small woody debris, boulders, undercut banks, etc?
Maintenance	If required, provide maintenance recommendations.
Recommendations	General recommendations for follow up. Could include revegetation, addition of substrate, fish sampling, etc.

4 Results and Discussion

4.1 Site Assessment Data

A summary of fish passage assessment procedures conducted through SERNbc since 2020 — including assessments, habitat confirmations, design, remediation, monitoring, fish sampling, eDNA sampling, and drone imagery — is provided in [Appendix - Assessment Data Summary \(page 51\)](#).

Orthoimagery and elevation model rasters generated since 2020 are catalogued in [Appendix - UAV Imagery \(page 53\)](#).

4.2 Planning — Habitat and Connectivity Modelling

4.2.1 Habitat Modelling

Habitat modelling from `bcfishpass` including access model, linear spawning/rearing habitat model and lateral habitat connectivity models for watershed groups within our study area were updated for the spring of 2025 and are included spatially in the collaborative GIS project. A snapshot of these outputs related to each modeled and PSCIS stream crossing structure are also included within an `sqlite` database within this year's project reporting/code repository [here](#).

4.2.1.1 Statistical Support for `bcfishpass` Fish Habitat Modelling Updates

Initial mapping of stream discharge and temperature causal effects pathways for the future purpose of focusing aquatic restoration actions in areas of highest potential for positive impacts on fisheries values (ie. elimination of areas from intrinsic models where water temperatures are likely too cold to support fish production) are detailed in Hill et al. (2024) which is included as [Attachment - Water Temperature Modelling \(page 81\)](#).

4.3 Fish Passage Assessments

Field assessments were conducted between September 29, 2025 and October 04, 2025 by Allan Irvine, R.P.Bio. The 2025 field season was directed at effectiveness monitoring and environmental DNA sampling rather than at new fish passage assessments. A total of 1 crossings were assessed, all of them reassessments of structures already inventoried in PSCIS. No Phase 1 assessments were completed.

Reassessments update the provincial record where a structure has been replaced or remediated, or where the existing PSCIS entry no longer reflects conditions at the crossing. They do not add new crossings to the inventory, so they do not carry the cost estimates and priority rankings that

accompany Phase 1 results. The amalgamated assessment record for the study area, spanning 2020 to 2025, is presented in [Appendix - Assessment Data Summary \(page 51\)](#).

4.4 Habitat Confirmation Assessments

No Phase 2 habitat confirmation assessments were conducted in 2025. Habitat confirmations characterise the quantity and quality of habitat upstream and downstream of a crossing prioritised during Phase 1 assessment; with no Phase 1 work completed this season, none were scheduled.

Habitat confirmation results from previous years in the Bulkley River, Kispiox River, Kalum River, Morice River, and Zymoetz River watershed groups are carried in the amalgamated record in [Appendix - Assessment Data Summary \(page 51\)](#), and in the reporting for those years.

Environmental DNA (eDNA) Sampling

New in 2025, environmental DNA (eDNA) sampling was incorporated into the program at both previously assessed and newly added sites. A total of 20 collection events were undertaken across 11 streams, comprising 19 real environmental samples, 1 field blanks, and 0 office blanks collected for quality assurance.

Per-species detection results across the 19 real environmental samples are summarized in Table 4.1. Detections use the ddPCR call threshold of ≥ 4 positive droplets applied by the laboratory. Results of 2-3 droplets were retested; where the additional droplets did not bring the total to 4 or more the sample was called negative. Site-level detail — the per-site detection table, field blanks, and retests — is provided in [Appendix - Environmental DNA Results \(page 57\)](#), and the [interactive eDNA map](#) shows toggleable per-species layers.

Table 4.1: Per-species summary of eDNA results across real environmental samples in the project area (excludes field and office blanks).

Species	Code	Sites tested	Detected (≥ 4 droplets)	Not detected
Rainbow Trout	RAIN	13	9	4
Bull Trout	BULT	18	7	11
Coho Salmon	COHO	19	3	16
Chinook Salmon	CHIN	17	0	17
Sockeye Salmon	SOCK	9	0	9

4.5 Engineering Design

No new designs were commissioned in 2025 due to uncertainty related to forest harvesting activities and lack of funding for the 50% costs of replacing structures. All sites with a design for remediation of fish passage can be found online within the Assessment Data Summary appendix of the report found [here](#).

4.6 Monitoring

In 2025, baseline or follow up monitoring data was gathered through completion of an effectiveness monitoring form (sites where remediation has been completed) and/or through eDNA sampling and acquisition of aerial imagery at the following sites (Table 4.2:

- Effectiveness monitoring was conducted on Peacock Creek (PSCIS crossing 197962) on the Morice FSR, where the oval culvert was replaced with a bridge in 2021. Environmental DNA sampling upstream and downstream of the crossing confirmed coho, rainbow trout and bull trout in the reach. Results are presented in the [Peacock Creek - 197962 - Monitoring Appendix \(page 61\)](#).
- Effectiveness monitoring was conducted on Waterfall Creek (PSCIS crossing 124421) at 11th Avenue, where the culvert was removed by the Gitksan Watershed Authorities. Riprap at the site has been buried and seeded, with good regrowth of grasses established and shrub planting planned.

Table 4.2: Summary of monitoring sites.

PSCIS ID	Stream	Road	Crossing type	Crossing subtype	Diameter or span (m)	Length or width (m)	Assessment comments
197962	Peacock Creek	Morice FSR	Open Bottom Structure	BRIDGE	20	10	Bridge installed in 2020 or 2021. Bridge span estimated from photos.
124421	Waterfall Creek	11th Avenue	Open Bottom Structure	FORD	–	–	Culvert was removed by Gitksan Watershed Authorities. Site riprap was buried with dirt and seeded with good regrowth of grasses. Shrub planting is planned for this fall.

4.7 Collaborative GIS Environment

Fieldwork activities and background spatial layers are maintained in a collaborative QGIS and Merpin Maps environment (sern_skeena_2023), which field crews and partners use to view, edit and add data. The layers loaded at the time of writing are listed in [Appendix - Collaborative GIS Layers \(page 55\)](#).

5 Recommendations

A major challenge in advancing fish passage restoration is the complexity of working across jurisdictions and with multiple stakeholders—rail and highway authorities, forestry ministries, licensees, and private landowners. These partners are often being asked to accommodate priorities that originate outside their mandates and budgets. Convincing them to invest in difficult, high-cost interventions—like modifying crossings or relocating infrastructure—requires navigating uncertainty about costs and ecological outcomes, as well as a disconnect between the benefits to watershed health and the internal pressures or performance goals of these agencies. It's a tough ask: to take on massive, uncertain projects when they're already stretched thin with their own responsibilities.

Fish passage restoration across British Columbia is further complicated by the legacy of infrastructure deeply embedded in the landscape. Roads, railways, highways, community infrastructure and private assets often constrain floodplains and disrupt natural hydrological processes. While targeted repairs to individual barriers are essential, they won't resolve the broader systemic issues without rethinking and restructuring how infrastructure interacts with watershed function. Loss of riparian vegetation and intensive beaver management only add to the degradation. Addressing these challenges means making strategic, well-communicated choices, building trust, and staying committed to a longer-term transformation.

A single definitive ranking of remediation priorities is not achievable, and the reason is worth stating plainly: the criteria that determine whether a crossing is worth remediating — species present and species possible, habitat quantity and quality, barrier severity, floodplain condition, constructability, and cost — are not measured in common units, so any ordering depends on which criterion is allowed to lead. The deciding constraint sits outside the project entirely, in the capacity and willingness of infrastructure owners and tenure holders to support implementation, both financially and across multi-year timelines. What can be done systematically is to maintain a well-characterised set of candidate sites, so that when a tenure holder is in a position to act, the ecological value of the opportunity and the appropriate scope of work are already understood.

Government, community groups, landowners, non-profits, industry and other stakeholders should work collaboratively to address the high and moderate priority barriers identified online within the report found [here](#). Progress on any front is meaningful, and aiming to remediate at least one high-priority site per year per watershed group—regardless of its overall rank—is a practical and effective approach.

To advance fish passage restoration in the Skeena region:

Effectiveness monitoring and building the state of knowledge

5 Recommendations

- Continue the effectiveness monitoring program at PSCIS crossings 197962 (Peacock Creek, Morice FSR) and 124421 (Waterfall Creek, 11th Avenue). Monitoring is where the program learns whether a remediation delivered what it was designed to deliver, and that is only visible over multiple years. Documenting an outcome that did not hold is as valuable as documenting one that did.
- Extend monitoring to additional remediated sites across the region so the record spans a range of structure types, treatment approaches and stream sizes rather than a small number of long-running sites.
- Share monitoring outcomes openly with tenure holders, regulators, First Nations and other restoration practitioners, so decisions elsewhere are informed by what has and has not worked here.
- Continue environmental DNA sampling as an annual, region-wide data layer. Sustained annual sampling is what turns single-year detections into distribution and trend information. The 2025 results already establish species presence at sites carrying no habitat confirmation assessment.
- Continue to develop a cost-effective monitoring framework that ties baseline and post-remediation data to measurable productivity gains from improved passage.

Partnerships, capacity and adaptive management

- Continue working with First Nations across the study area — Wet'suwet'en, Gitksan and Tsimshian — on site selection, monitoring, and interpretation of what the data show.
- Continue partnerships with stewardship groups and educational institutions, including the University of Northern British Columbia, so field programs double as training and research opportunities and the regional knowledge base grows beyond any single project.
- Maintain partnerships with rail, highway and forestry tenure holders, and with the Habitat Conservation Trust Foundation and the Ministry of Transportation and Infrastructure, to support funding, site selection, remediation and monitoring.
- Feed monitoring results back into prioritization and design. The purpose of a monitoring program is adaptive management, and what is learned at Peacock Creek and Waterfall Creek should change which treatments are proposed elsewhere.

Watershed-scale context — floodplain health and climate departure

- Map functional floodplain extent for the Bulkley River watershed group, then extend to Morice, Zymoetz and Kispiox. Passage and floodplain health are the same problem viewed from two angles: reconnecting a stream longitudinally achieves less where the floodplain is severed laterally. Starting with one watershed group also allows the delivery route — publishing to a spatial catalogue that the report reads from, rather than committing rasters to the repository — to be proven before it is repeated.

- Extend the climate departure analysis to the study area watershed groups, so climate trajectory sits alongside barrier status wherever prioritization is done.
- Build capacity in partner organizations to interpret and use the floodplain and climate departure layers, so the opportunities and constraints these analyses surface are visible to the people whose decisions shape watershed-scale outcomes.

Environmental DNA — resolving what the detections mean

- Follow up the multi-species detections at Peacock Creek, where coho, rainbow trout and bull trout were all confirmed, and at Corya Creek and Sandstone Creek. These are the sites where eDNA has established that the habitat is occupied and where passage condition therefore carries the clearest consequence.
- Carry environmental DNA results into prioritization rather than reporting them alongside it. The detections establish species presence at sites with no habitat confirmation assessment, and that belongs in the ranking.

Prioritization and modelling integration

- Continue integrating climate and temperature modelling into prioritization, in both directions. Prioritizing crossings that open access to cold, drought-resistant habitat addresses reaches at the warm edge of their thermal range. The same data answers the opposite question: modelling growing season degree days identifies reaches currently too cold to support production where warming may be making production possible, meaning the restoration value of a barrier below such a reach can be rising rather than falling.
- Calibrate intrinsic habitat models against areas of known high-value spawning and rearing identified through fish presence observations, quantifying model uncertainty and refining prioritization.
- Resume Phase 1 fish passage assessments in the study area. 2025 added two reassessments and no new inventory; the candidate set that prioritization draws on only grows through Phase 1 work.

Data infrastructure and open access

- Develop a public webmap interface to communicate fish passage priorities, baseline conditions, monitoring results and partner activities to collaborators, First Nations, industry, regulators and the public.
- Extend the open-access pattern to environmental DNA data, making detections, samples, methods and metadata queryable through a shared interface rather than scattered across project repositories.
- Support a community of practice around regional water temperature and fish presence data with the University of Northern British Columbia, partner organizations and other researchers, focused on collective capacity to ask and answer regional questions rather than every program reconstructing the same picture from fragmented sources.

Appendix - Salmon Stock Assessment Data

The stock assessment extract has not yet been regenerated for this region. The previous extract covered a different study area and was removed rather than published here. See the repository issues for the regeneration task.

Fisheries and Oceans Canada stock assessment data was accessed via the [NuSEDS-New Salmon Escapement Database System](#) through the [Open Government Portal](#) with results presented online in this appendix at [app-stock-assessment.html](#) (Fisheries and Oceans Canada, n.d.). A brief memo on the data extraction process is available [here](#).

Appendix - Fish Species by Watershed Group

Historical fish observations recorded in the watershed groups covered by this project (MoE 2024a). The print version collapses presence to a single column listing the watershed groups each species has been recorded in; the online version gives one column per watershed group.

Table 5.1: Fish species recorded in the Bulkley, Kispiox, Kalum, Morice, and Zymoetz watershed groups.

Scientific Name	Species Name	BC List	COSEWIC	Present in WSGs
<i>Catostomus catostomus</i>	Longnose Sucker	Yellow	–	Bulkley, Kispiox, Morice, Zymoetz
<i>Catostomus commersonii</i>	White Sucker	Yellow	–	Bulkley, Kalum, Kispiox, Morice
<i>Catostomus macrocheilus</i>	Largescale Sucker	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Chrosomus eos</i>	Northern Redbelly Dace	Yellow	–	Bulkley
<i>Coregonus clupeaformis</i>	Lake Whitefish	Yellow	–	Bulkley, Kispiox, Morice
<i>Coregonus sardinella</i>	Least Cisco	Blue	–	Kalum
<i>Cottus aleuticus</i>	Coastrange Sculpin (formerly Aleutian Sculpin)	Yellow	–	Bulkley, Kalum, Kispiox, Morice
<i>Cottus asper</i>	Prickly Sculpin	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Cottus cognatus</i>	Slimy Sculpin	Yellow	–	Kalum, Kispiox
<i>Couesius plumbeus</i>	Lake Chub	Yellow	DD	Bulkley, Kalum, Kispiox, Morice
<i>Entosphenus tridentatus</i>	Pacific Lamprey	Yellow	–	Bulkley, Kalum, Morice
<i>Gasterosteus aculeatus</i>	Threespine Stickleback	Yellow	–	Kalum, Kispiox
<i>Hybognathus hankinsoni</i>	Brassy Minnow	No Status	–	Bulkley
<i>Lampetra ayresii</i>	River Lamprey	Yellow	–	Kalum
<i>Lota lota</i>	Burbot	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Mylocheilus caurinus</i>	Peamouth Chub	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Oncorhynchus clarkii</i>	Cutthroat Trout	No Status	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Oncorhynchus clarkii</i>	Cutthroat Trout (Anadromous)	No Status	–	Bulkley, Kispiox, Zymoetz
<i>Oncorhynchus clarkii clarkii</i>	Coastal Cutthroat Trout	Blue	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Oncorhynchus clarkii lewisi</i>	Westslope (Yellowstone) Cutthroat Trout	Blue	SC (Nov 2016)	Kalum, Kispiox

Appendix - Fish Species by Watershe...

Scientific Name	Species Name	BC List	COSEWIC	Present in WSGs
<i>Oncorhynchus gorbuscha</i>	Pink Salmon	Not Reviewed	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Oncorhynchus keta</i>	Chum Salmon	Not Reviewed	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Oncorhynchus kisutch</i>	Coho Salmon	Not Reviewed	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Oncorhynchus mykiss</i>	Rainbow Trout	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Oncorhynchus mykiss</i>	Steelhead	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Oncorhynchus mykiss</i>	Steelhead (Summer-run)	Yellow	–	Bulkley, Morice
<i>Oncorhynchus mykiss</i>	Steelhead (Winter-run)	Yellow	–	Kispiox, Zymoetz
<i>Oncorhynchus nerka</i>	Kokanee	Not Reviewed	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Oncorhynchus nerka</i>	Sockeye Salmon	Not Reviewed	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Oncorhynchus tshawytscha</i>	Chinook Salmon	Not Reviewed	E/T/SC/DD/NAR (Nov 2020)	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Prosopium coulterii</i>	Pygmy Whitefish	Yellow	NAR (Nov 2016)	Bulkley, Kispiox, Morice
<i>Prosopium coulterii</i> pop. 3	Giant Pygmy Whitefish	Yellow	NAR (Nov 2016)	Bulkley
<i>Prosopium cylindraceum</i>	Round Whitefish	Yellow	–	Kalum
<i>Prosopium williamsoni</i>	Mountain Whitefish	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Ptychocheilus oregonensis</i>	Northern Pikeminnow	Yellow	–	Bulkley, Kispiox, Morice, Zymoetz
<i>Pungitius pungitius</i>	Ninespine Stickleback	Unknown	–	Bulkley
<i>Rhinichthys cataractae</i>	Longnose Dace	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Rhinichthys falcatus</i>	Leopard Dace	Yellow	NAR (May 1990)	Morice
<i>Richardsonius balteatus</i>	Redside Shiner	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Salvelinus confluentus</i>	Bull Trout	Blue	SC (Nov 2012)	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Salvelinus fontinalis</i>	Brook Trout	Exotic	–	Bulkley, Morice
<i>Salvelinus malma</i>	Dolly Varden	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
<i>Salvelinus namaycush</i>	Lake Trout	Yellow	–	Bulkley, Kispiox, Morice
–	All Salmon	–	–	Kalum, Kispiox
–	Arctic Char	–	–	Morice
–	Chub (General)	–	–	Kispiox

Scientific Name	Species Name	BC List	COSEWIC	Present in WSGs
–	Cutthroat/Rainbow cross	–	–	Bulkley, Kalum, Kispiox
–	Dace (General)	–	–	Morice
–	Lamprey (General)	–	–	Bulkley, Kalum, Kispiox, Morice
–	Minnow (General)	–	–	Bulkley, Kispiox, Morice
–	Mottled Sculpin	–	–	Bulkley
–	Salmon (General)	–	–	Bulkley, Kispiox, Morice, Zymoetz
–	Sculpin (General)	–	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
–	Squanga	–	–	Kispiox
–	Stickleback (General)	–	–	Kalum, Kispiox
–	Sucker (General)	–	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
–	Verified DV BT hybrid	–	–	Kalum
–	Whitefish (General)	–	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
* COSEWIC abbreviations : SC - Special concern DD - Data deficient NAR - Not at risk E - Endangered T - Threatened BC List definitions : Yellow - Species that is apparently secure Blue - Species that is of special concern Exotic - Species that have been moved beyond their natural range as a result of human activity				

Appendix - Assessment Data Summary

Fish passage assessment procedures conducted through SERNbc since 2020 are amalgamated here — assessments, habitat confirmations, design, remediation, monitoring, fish sampling, eDNA sampling and drone imagery. The detailed per-site table is interactive and is provided online within this appendix [here](#).

Table 5.2:
Summary
of fish
passage
assessment
procedures
conducted
in northern
British
Columbia
through
SERNbc.

**NOTE: To
view all
columns in
the table -
please
click on
one of the
sort
arrows
within
column
headers
before
scrolling
to the
right.**

Table 5.3:
Details of
fish
passage
assessment
procedures
conducted
in northern
British
Columbia
through
SERNbc.

Appendix - UAV Imagery

Since 2020, orthoimagery and elevation model rasters have been generated and stored as Cloud Optimized Geotiffs on a cloud service provider (AWS) with select imagery linked to in the collaborative GIS project. Additionally - a tile service has been set up to facilitate viewing and downloading of individual images, provided

[athttps://www.newgraphenvironment.com/fish_passage_skeena_2025_reporting//results-and-discussion.html](https://www.newgraphenvironment.com/fish_passage_skeena_2025_reporting//results-and-discussion.html).

Appendix - Collaborative GIS Layers

In addition to numerous layers documenting fieldwork activities since 2020, a summary of background information spatial layers and tables loaded to the collaborative GIS project (sern_skeena_2023) at the time of writing (2026-08-04) are listed below. The layer table is interactive and is provided online within this appendix [here](#).

Appendix - Environmental DNA Results

This appendix carries the site-level detail behind the headline counts in the Results chapter. Field blanks are reported separately because they test the sampling protocol rather than a location. Office blanks are excluded entirely — they are distilled-water controls filtered at the end of a session at accommodation, so although they carry real GPS coordinates, those coordinates record where the filtering happened rather than a stream site. The interactive map is linked at the end.

Per-site detection results

Format: CODE(max_droplets), with * flagging samples that UNBC reran. Detected = ≥ 4 positive droplets, the call threshold applied by the laboratory. Everything below that threshold is reported as not detected, including 2-3 droplet results that were retested without reaching the cutoff.

Table 5.4: Site-level eDNA detection results across real environmental samples in the project area. UNBC ID is the lab sample identifier; Pos. control = yes indicates the sample was collected at a site where the target species was also confirmed by electrofishing.

Site ID	Stream	UNBC ID	Pos. control	Detected	Not detected
197360_ds_ed	Riddeck Creek	S61182		RAIN(36)*	BULT(0), CHIN(1), COHO(1)*
197378_Us_ed1	Tributary To Owen Creek	S61184		RAIN(512)	BULT(0), CHIN(0), COHO(1)
197378_ds_ed1b	Tributary To Owen Creek	S61185		RAIN(679)	BULT(0), CHIN(0), COHO(0)
197379_ds_ed	Avin Creek	S61186		RAIN(30)	BULT(0), CHIN(0), COHO(0)
197379_us_ed1	Avin Creek	S31493		RAIN(111)	BULT(0), COHO(1), SOCK(0)
197960_ds_ed1	Corya Creek	S50879		BULT(113), COHO(12)*	CHIN(0), SOCK(0)
197962_ds_ed	Peacock Creek	S60678		COHO(8), RAIN(124)	CHIN(0), SOCK(0)
197962_us_ed	Peacock Creek	S60880		BULT(84), COHO(6)	CHIN(0), SOCK(0)
58067_ds_ed1	Tributary To Gramophone Creek	S61187		RAIN(96)	BULT(0), CHIN(0), COHO(0)
8525_ds_ed1	Tributary To Coal Creek	S61188		BULT(128)	CHIN(0), COHO(0), RAIN(0)
8525_us_ed1	Tributary To Coal Creek	S61189	yes	BULT(110)	CHIN(0), COHO(0), RAIN(0)
8525_us_ed2	Tributary To Coal Creek	S61190		BULT(18)	CHIN(0), COHO(0), RAIN(0)
8530_ds_ed1	Sandstone Creek	S61191		BULT(63), RAIN(59)	CHIN(0), COHO(0)
8530_us_ed1	Sandstone Creek	S61192		BULT(66), RAIN(57)	CHIN(0), COHO(0)
8547_ds_ed1	Tributary To McDonnell Lake	S60881		—	BULT(2)*, CHIN(0), COHO(0), SOCK(0)
8547_ds_ed2	Tributary To McDonnell Lake	S60377		—	BULT(0), COHO(0), RAIN(0), SOCK(0)

Appendix - Environmental DNA Results

Site ID	Stream	UNBC ID	Pos. control	Detected	Not detected
8547_us_ed1	Tributary To McDonnell Lake	S60274		—	BULT(0), CHIN(0), COHO(0), SOCK(0)
bulkley_falls_ds_ed1	Bulkley River	S60275		—	BULT(0), CHIN(0), COHO(0), SOCK(0)
bulkley_falls_ds_ed2	Bulkley River	S60276		—	BULT(1), CHIN(0), COHO(1)*, SOCK(0)

Field blanks

Field blanks filter distilled water at the field site to test for protocol contamination during collection and processing. **Detections in blanks indicate potential cross-contamination, not species presence at the location** — these rows are presented for QA transparency only, not as eDNA detections.

Table 5.5: Field blanks collected during the `r_params$project_year` eDNA program in the project area. Detections in blanks reflect protocol-side contamination, not site eDNA.

Site ID	Stream	UNBC ID	Detected	Not detected
197378_ds_ed1a	—	S61183	—	BULT(0), CHIN(0), COHO(0), RAIN(0)

Retests

UNBC reran a subset of samples to confirm borderline results. The table below lists every site x species combination that was rerun, including reruns of both real environmental samples and field blanks.

Table 5.6: UNBC retests for the `r_params$project_year` eDNA samples. Runs is the total number of times the sample x species combination was assayed; Max droplets is the highest positive-droplet count observed across all runs.

Site ID	Stream	Species	Code	Sample type	Runs	Max droplets	Result
197360_ds_ed	Riddeck Creek	Bull Trout	BULT	environmental	6	0	Not detected
197360_ds_ed	Riddeck Creek	Chinook Salmon	CHIN	environmental	6	1	Not detected
197360_ds_ed	Riddeck Creek	Coho Salmon	COHO	environmental	6	1	Not detected
197360_ds_ed	Riddeck Creek	Rainbow Trout	RAIN	environmental	6	36	Detected (≥4 droplets)
197378_ds_ed1a	—	Bull Trout	BULT	field blank	4	0	Not detected
197378_ds_ed1a	—	Chinook Salmon	CHIN	field blank	4	0	Not detected
197378_ds_ed1a	—	Coho Salmon	COHO	field blank	4	0	Not detected

Interactive map

Site ID	Stream	Species	Code	Sample type	Runs	Max droplets	Result
197960_ds_ed1	Corya Creek	Coho Salmon	COHO	environmental	4	12	Detected (≥ 4 droplets)
8547_ds_ed1	Tributary To McDonnell Lake	Bull Trout	BULT	environmental	4	2	Not detected
bulkley_falls_ds_ed2	Bulkley River	Coho Salmon	COHO	environmental	4	1	Not detected

Interactive map

The interactive eDNA map shows site-level detection results with toggleable per-species layers and an opt-in Controls layer for field blanks. See the [interactive eDNA map](#).

Peacock Creek - 197962 - Monitoring Appendix

PSCIS crossing 197962 on Peacock Creek is located on the Morice Forest Service Road within the Morice River watershed group. The culvert at this crossing was replaced with a bridge in 2021, and the site was revisited in 2025 under the effectiveness monitoring program.

Crossing Characteristics

Before replacement, the structure was a corrugated metal culvert with a concrete headwall, ranked a barrier to upstream migration under the provincial protocol (MoE 2011). The assessment recorded a partial inlet drop of varying height caused by large woody debris and boulders. Photographs taken during the September 2021 assessment show the outlet perched above a plunge pool, with the drop and the pool visible below the headwall (Figure [5.1](#)).

The replacement structure is an open-span bridge. Photographs from the 2025 monitoring visit show an unconstricted channel through the crossing, with the bed continuous through the opening and no residual outlet drop (Figure [5.2](#)).



Figure 5.1: Culvert on Peacock Creek at PSCIS crossing 197962 on the Morice FSR, photographed on 2021-09-07 before replacement. The outlet is perched above a plunge pool.

Crossing Characteristics



Figure 5.2: Bridge on Peacock Creek at PSCIS crossing 197962 on the Morice FSR, photographed on 2025-09-29 following replacement.



Figure 5.3: Substrate condition through the crossing at PSCIS crossing 197962, recorded during the 2025 effectiveness monitoring visit.

Habitat and Fish Presence

Habitat confirmation assessments were completed at this site in September 2021, before the culvert was replaced. Both the reach upstream and the reach downstream of the crossing were rated high habitat value, with 750 m surveyed upstream and 600 m downstream. The upstream reach was described as complex habitat with undercut banks, large and small woody debris and pools; the downstream reach as carrying consistent undercut banks with woody debris and overhanging vegetation providing cover. Approximately 4.8 km of upstream habitat length was estimated, and the site was ranked a high priority for remediation.

The site was electrofished on 2021-09-17 at three closed-site multi-pass locations upstream of the crossing and three downstream, capturing 96 fish across four species (Table [5.7](#)). Dolly Varden and rainbow trout were caught on both sides of the crossing. Coho salmon and cutthroat trout were caught only downstream. Fork lengths ranged from 48 to 148 mm across all species, with coho between 63 and 85 mm.

Discussion

Table 5.7: Fish captured by multi-pass closed-site electrofishing at PSCIS crossing 197962 on 2021-09-17, before the culvert was replaced. Three sites upstream and three downstream.

Species	Downstream	Upstream	Total
Coho Salmon	10	0	10
Cutthroat Trout	1	0	1
Dolly Varden	16	40	56
Rainbow Trout	26	3	29
Total	53	43	96

Replacement of the culvert began three days later, on 2021-09-20, so the entire 2021 record — assessment, habitat confirmation and fish sampling — describes the site in its pre-remediation condition.

In 2025, environmental DNA was collected both upstream and downstream of the crossing. Downstream, rainbow trout and coho were detected. Upstream, bull trout and coho were detected. Chinook and sockeye were not detected at either location (Table [5.8](#)).

Table 5.8: Environmental DNA detection results at PSCIS crossing 197962, upstream and downstream of the crossing.

Site ID	UNBC ID	Detected	Not detected
197962_ds_ed	S60678	COHO(8), RAIN(124)	CHIN(0), SOCK(0)
197962_us_ed	S60880	BULT(84), COHO(6)	CHIN(0), SOCK(0)

Discussion

The site carries an uncommonly strong record. The pre-remediation habitat confirmation rated both reaches high value and estimated 4.8 km of habitat upstream, multi-pass electrofishing established what was present on each side of the culvert before it was replaced, and the 2025 environmental DNA re-establishes species presence four years after replacement.

Coho and cutthroat trout were captured only downstream of the culvert in 2021. In 2025, coho was detected on both sides of the bridge.

Recommendations for continued monitoring at this site:

- Resample for environmental DNA at both locations across multiple points in the season. The 2025 samples were collected on one day at the end of September.
- Assay the same species panel at both locations so the two are directly comparable. In 2025 the upstream and downstream samples were assayed for different sets of targets.
- Repeat the electrofishing conducted in 2021 at the same closed-site locations. Pre-remediation multi-pass data exist for three sites upstream and three downstream, which is an uncommonly strong baseline and the most direct available measure of whether the replacement changed fish use rather than only passage condition.
- Continue photographic monitoring of substrate, cover, bank stability and revegetation through the crossing, so channel recovery following construction can be tracked over time.

References

- Amazon Web Services. 2025. *Aws/Aws-Cli: Universal Command Line Interface for Amazon Web Services*. V. 2.18.15. Released. <https://github.com/aws/aws-cli>.
- Bell, M. C. 1991. *Fisheries Handbook of Engineering Requirements and Biological Criteria*. https://www.fs.fed.us/biology/nsaec/fishxing/fplibrary/Bell_1991_Fisheries_handbook_of_engineering_requirements_and.pdf.
- Bourne, Christina, Dan Kehler, Yolanda Wiersma, and David Cote. 2011. "Barriers to Fish Passage and Barriers to Fish Passage Assessments: The Impact of Assessment Methods and Assumptions on Barrier Identification and Quantification of Watershed Connectivity." *Aquatic Ecology* 45: 389–403. <https://doi.org/10.1007/s10452-011-9362-z>.
- Bramblett, Robert G., Mason D. Bryant, Brenda E. Wright, and Robert G. White. 2002. "Seasonal Use of Small Tributary and Main-Stem Habitats by Juvenile Steelhead, Coho Salmon, and Dolly Varden in a Southeastern Alaska Drainage Basin." *Transactions of the American Fisheries Society* 131 (3): 498–506. [https://doi.org/10.1577/1548-8659\(2002\)131<0498:SUOSTA>2.0.CO;2](https://doi.org/10.1577/1548-8659(2002)131<0498:SUOSTA>2.0.CO;2).
- Busch, D.Shallin, Mindi Sheer, Kelly Burnett, Paul McElhany, and Tom Cooney. 2011. "Landscape-Level Model to Predict Spawning Habitat For Lower Columbia River Fall Chinook Salmon (*Oncorhynchus Tshawytscha*): Intrinsic Potential Model for Spawning Fall Chinook Salmon." *River Research and Applications* 29 (3): 297–312. <https://doi.org/10.1002/rra.1597>.
- Bustard, D, and C Schell. 2002. *Conserving Morice Watershed Fish Populations and Their Habitat*. Community Futures Development Corporation of Nadina. <https://waves-vagues.dfo-mpo.gc.ca/Library/315091.pdf>.
- Calvin, Katherine, Dipak Dasgupta, Gerhard Krinner, et al. 2023. *IPCC: Summary for Policymakers*. In: *Climate Change 2023: Synthesis Report*. First. Intergovernmental Panel on Climate Change (IPCC). <https://doi.org/10.59327/IPCC/AR6-9789291691647>.
- Canada, Environment and Climate Change. 2024. "National Water Data Archive: HYDAT." Service description. Aem. <https://www.canada.ca/en/environment-climate-change/services/water-overview/quantity/monitoring/survey/data-products-services/national-archive-hydat.html>.
- Clarkin, K, A Connor, M Furniss, et al. 2005. *National Inventory and Assessment Procedure For Identifying Barriers to Aquatic Organism Passage at Road-Stream Crossings*. United States Department of Agriculture, Forest Service, National Technology; Development Program. <https://www.fs.fed.us/biology/nsaec/fishxing/publications/PDFs/NIAP.pdf>.
- cogeotiff. (2018) 2025. *Cogeotiff/Rio-Cogeo*. March 9; Cogeotiff, released March 19. <https://github.com/cogeotiff/rio-cogeo>.
- Cooney, Thomas, and Damon Holzer. 2006. *Appendix C: Interior Columbia Basin Stream Type Chinook Salmon and Steelhead Populations: Habitat Intrinsic Potential Analysis*. 21.
- Cote, David, P Frampton, M Langdon, and R Collier. 2005. *Fish Passage and Stream Habitat Restoration in Terra Nova National Park Highway Culverts*.
- Department of Fisheries and Oceans. 1991. *Fish Habitat Inventory and Information Program SISS Stream Summary Catalogue. Subdistrict 4D, Smithers (Volume 2)*. Bulkley.<https://waves-vagues.dfo-mpo.gc.ca/library-bibliotheque/40602369.pdf>.
- Development Seed. (2019) 2025. *Developmentseed/Titiler*. V. 0.21.1. June 28; Development Seed, released. <https://github.com/developmentseed/titiler>.
- Diebel, M. W., M. Fedora, S. Cogswell, and J. R. O'Hanley. 2015. "Effects of Road Crossings on Habitat Connectivity for Stream-Resident Fish: STREAM-RESIDENT FISH HABITAT CONNECTIVITY." *River Research and Applications* 31 (10): 1251–61. <https://doi.org/10.1002/rra.2822>.

References

- Dyson, J. B. 1949. *Bulkley Falls Investigation Report*. <https://data.skeenasalmon.info/dataset/0af0ecf8-0d55-4d48-9bde-d869db0fb71a/resource/b3360add-f7df-4906-8cb5-c2cb4a3e7fa9/download/bulkley-falls-investigation-dfo-1949.pdf>.
- ECCC. 2016. *Climate Data and Scenarios for Canada: Synthesis of Recent Observation and Modelling Results*. Environment; Climate Change Canada (ECCC). https://publications.gc.ca/collections/collection_2016/eccc/En84-132-2016-eng.pdf.
- Elmer, Laura. 2021. *Initial Assessment of Fish Habitat Usage and Accessibility in Clear Creek Eastern Side Channel*. <https://a100.gov.bc.ca/pub/acat/public/viewReport.do?reportId=61024>.
- Fernandes, I. M., Y. F. Bastos, D. S. Barreto, L. S. Lourenço, and J. M. Penha. 2017. "The Efficacy of Clove Oil as an Anaesthetic and in Euthanasia Procedure for Small-Sized Tropical Fishes." *Brazilian Journal of Biology = Revista Brasileira De Biologia* 77 (3): 444–50. <https://doi.org/10.1590/1519-6984.15015>.
- Finnegan, Noah J., Gerard Roe, David R. Montgomery, and Bernard Hallet. 2005. "Controls on the Channel Width of Rivers: Implications for Modeling Fluvial Incision of Bedrock." *Geology* 33 (3): 229–32. <https://doi.org/10.1130/G21171.1>.
- Fish Passage Technical Working Group. 2011. *A Checklist for Fish Habitat Confirmation Prior to the Rehabilitation Fo a Stream Crossing*. <https://www2.gov.bc.ca/assets/gov/environment/natural-resource-stewardship/land-based-investment/forests-for-tomorrow/checklist-for-fish-habitat-confirmation-201112.pdf>.
- Fisheries and Oceans Canada. n.d. "NuSEDS-New Salmon Escapement Database System - Open Government Portal." Accessed April 17, 2025. <https://open.canada.ca/data/en/dataset/c48669a3-045b-400d-b730-48aaf8c5ee6>.
- FLNRO. 2013a. *Bulkley River Angling Management Plan*. Ministry of Forests, Lands; Natural Resource Operations (FLNRO). <https://www2.gov.bc.ca/assets/gov/environment/plants-animals-and-ecosystems/fish-fish-habitat/fishery-resources/region-6-skeena/skeena-amp-bulkley-river-2013.pdf>.
- FLNRO. 2013b. *Overview of Angling Management Plans for the Skeena Watershed*. Ministry of Forests, Lands, Natural Resource Operations (FLNRO). <https://www2.gov.bc.ca/assets/gov/environment/plants-animals-and-ecosystems/fish-fish-habitat/fishery-resources/region-6-skeena/skeena-amp-overview-2013.pdf>.
- FLNRORD. 2013. *Zymoetz River Class I Section Angling Management Plan*. <https://www.env.gov.bc.ca/skeena/fish/AMPs/Zymoetz1 AMP.pdf>.
- FLNRORD. 2017. *Natural Resource Stewardship Monitoring and Assessment Report for the Wet'suwet'en Hereditary Territory*. <https://www2.gov.bc.ca/assets/gov/farming-natural-resources-and-industry/forestry/integrated-monitoring/nrsmonitoringandassessmentreport-wetsuweten.pdf>.
- FLNRORD. 2019. *Freshwater Fishing Regulations Synopsis*. Ministry of Forests, Lands, Natural Resource Operations & Rural Development (FLNRORD). https://www2.gov.bc.ca/assets/gov/sports-recreation-arts-and-culture/outdoor-recreation/fishing-and-hunting/freshwater-fishing/region_6_skeena.pdf.
- FLNRORD. 2020a. "Digital Road Atlas (DRA) - Master Partially-Attributed Roads - Data Catalogue." Published by Ministry of Forests, Lands, Natural Resource Operations and Rural Development. <https://catalogue.data.gov.bc.ca/dataset/digital-road-atlas-dra-master-partially-attributed-roads>.
- FLNRORD. 2020b. "Forest Tenure Road Section Lines - Data Catalogue." Published by Ministry of Forests, Lands, Natural Resource Operations and Rural Development (FLNRORD). <https://catalogue.data.gov.bc.ca/dataset/forest-tenure-road-section-lines>.
- Forest Investment Account. 2003. *Guidelines for in-Stream and Off-Channel Routine Effectiveness Evaluation*. British Columbia Ministry of Forests. http://env.gov.bc.ca/wld/documents/fia_docs/ree_guidelines.pdf.

Gilbert, D. E., J. E. Morris, A. R. Kaveney, and J. D. Dery. 2022. "Sub-Hourly Water Temperature Data Collected Across the Nechako Watershed, 2019-2021 | Elsevier Enhanced Reader." <https://doi.org/10.1016/j.dib.2022.108425>.

"Gitxsan Huwilp Government." n.d. Accessed May 29, 2024. <https://gitxsan.ca/>.

Gottesfeld, Allen, Ken Rabnett, and Peter Hall. 2002. *Conserving Skeena Fish Populations and Their Habitat - Skeena Stage I Watershed-based Fish Sustainability Plan*. Skeena Fisheries Commission. <https://www.psf.ca/sites/default/files/Skeena%20WFSP%2012%20%28low%20res%29.pdf>.

Gottesfeld, A, and K Rabnett. 2007. *Skeena Fish Populations and Their Habitat*. Skeena Fisheries Commission.

"Groups Challenge Rush Start of Construction on Pipeline." 2025. Ecojustice, March 24. <https://ecojustice.ca/news/groups-in-court-to-challenge-the-bending-of-permit-requirements-to-rush-start-of-construction-on-900-km-pipeline/>.

Healthy Watersheds Initiative. 2021. "Riparian Restoration Surveys on Streams Near Terrace - Healthy Watersheds Initiative." June 8. <https://healthywatersheds.ca/project/riparian-restoration-surveys-on-streams-near-terrace/>.

Hill, N. H., J. L. Thorley, and A. Irvine. 2024. *Spatial Stream Network Analysis of Nechako Watershed Stream Temperatures 2022b*. Poisson Consulting Analysis Appendix. <https://www.poissonconsulting.ca/f/1295467017>.

Hobbs, Jared, Caren C. Helbing, and Caren Goldberg. 2025. *Environmental DNA Protocol for Freshwater Aquatic Ecosystems: Version 3.0*. BC Ministry of Environment, Ecosystems Branch.

IBM Business Consulting Services. 2006. "Valuation of the Wild Salmon Economy of the Skeena River Watershed." https://www.psf.ca/sites/default/files/IBM_skeena_report_061.pdf.

ILMB. 2007. *Morice Land and Resource Management Plan*. {Ministry of Agriculture; Lands - Integrated Land Management Bureau (ILMB). https://www2.gov.bc.ca/assets/gov/farming-natural-resources-and-industry/natural-resource-use/land-water-use/crown-land/land-use-plans-and-objectives/skeena-region/morice-lrmp/morice_lrmp_july2007.pdf.

Irvine. 2021. *Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning*. https://newgraphenvironment.github.io/fish_passage_bulkley_2020_reporting/.

Irvine. (2021) 2022. *Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning 2021*. https://github.com/NewGraphEnvironment/fish_passage_skeena_2021_reporting.

Irvine. (2025) 2025. *NewGraphEnvironment/Stac_uav_bc*. January 29, released. https://github.com/NewGraphEnvironment/stac_uav_bc.

Irvine, Al, and Lucy Schick. 2024. *Restoring Fish Passage in the Fraser Region - 2023*. New Graph Environment Ltd. Prepared for Ministry of Transportation and Infrastructure and Society for Ecosystem Restoration in Northern BC (SERNbc). https://newgraphenvironment.com/fish_passage_fraser_2023_reporting/.

Kemp, P. S., and J. R. O'Hanley. 2010. "Procedures for Evaluating and Prioritising the Removal of Fish Passage Barriers: A Synthesis: EVALUATION OF FISH PASSAGE BARRIERS." *Fisheries Management and Ecology*, no—. <https://doi.org/10.1111/j.1365-2400.2010.00751.x>.

Kirsch, J M, Joseph D Buckwalter, and Daniel J Reed. 2014. *Fish Inventory and Anadromous Cataloging in the Susitna River, Matanuska River, and Knik River Basins, 2003 and 2011*. 149.

Kitsumkalum Band. n.d. "Our Culture." Kitsumkalum, a Galts'ap (community) of the Tsimshian Nation. Accessed April 15, 2024. <https://kitsumkalum.com/about/our-culture/>.

"LNG Pipelines | Skeena Watershed Conservation Coalition." n.d. Accessed April 15, 2025. <https://skeenawatershed.com/lng-pipelines/>.

References

- McElhanney. 2022. *Kitsumkalum River Flood Mitigation Plan*. https://www.terrace.ca/sites/default/files/docs/Kitsumkalum%20River%20Flood%20Mitigation%20Plan_no%20appendices.pdf.
- MoE. 2011. *Field Assessment for Determining Fish Passage Status of Closed Bottom Structures*. BC Ministry of Environment (MoE). <https://www2.gov.bc.ca/assets/gov/environment/natural-resource-stewardship/land-based-investment/forests-for-tomorrow/field-assessment-for-determining-fish-passage-status-of-cbs.pdf>.
- MoE. 2024a. "Known BC Fish Observations and BC Fish Distributions." <https://catalogue.data.gov.bc.ca/dataset/known-bc-fish-observations-and-bc-fish-distributions>.
- MoE. 2024b. "Provincial Obstacles to Fish Passage - Data Catalogue." <https://catalogue.data.gov.bc.ca/dataset/provincial-obstacles-to-fish-passage>.
- Muñoz Sabater, Joaquín. 2019. *ERA5-Land Hourly Data from 1950 to Present*. Copernicus Climate Change Service (C3S) Climate Data Store (CDS). <https://doi.org/10.24381/cds.e2161bac>.
- Murray, Brent W., and Barry Booth. 2023. *Report on the 2023 eDNA Survey of Study Creeks in Nechako Watershed from Prince George to the Upper Fraser*. University of Northern British Columbia, Department of Ecosystem Science and Management. <https://www.newssociety.org/uploads/images/News%20report%20Images/e-DNA%20summary-%20UNBC-NEWSS%202023.pdf>.
- Norris, Simon. (2020) 2024. *Smnorris/Bcfishpass*. November 7, released January 12. <https://github.com/smnorris/bcfishpass>.
- Office of the Wet'suwet'en. 2013. *Wet'suwet'en Title and Rights Regarding Canada Department of Fisheries & Oceans And Pacific Trails Pipeline*. http://www.wetsuweten.com/files/PTP_FHCP_Response_to_DFO-25Nov13-Final.pdf.
- "Office of the Wet'suwet'en." 2021. <http://www.wetsuweten.com/>.
- Oliver, Allison. 2018. *Analysis of Water Quality Monitoring in the Morice Water Management Area*. <https://data.skeenasalmon.info/dataset/analysis-of-water-quality-monitoring-in-the-morice-water-management-area/resource/17125deb-57c3-4a91-b64c-377735bc2c32>.
- OpenDroneMap Authors. (2014) 2025. *OpenDroneMap/ODM*. V. 3.5.4. September 25; OpenDroneMap, released. <https://github.com/OpenDroneMap/ODM>.
- Pacific Climate Impacts Consortium. n.d. "Gridded Hydrologic Model Output." Accessed March 26, 2025. https://data.pacificclimate.org/portal/hydro_model_out/map/.
- Porter, Marc, Darcy Pickard, Katherine Wieckowski, and Katy Bryan. 2008. *Developing Fish Habitat Models for Broad-Scale Forest Planning in the Southern Interior of B.C.* ESSA Technologies Ltd. and B.C. Ministry of the Environment (MOE) for B.C. Forest Science Program. https://www.for.gov.bc.ca/hfd/library/FIA/2008/FSP_Y081231.pdf.
- Powell, J. V., Vickie D. Jensen, and Anne-Marie Pedersen. 2018. "Gitksan." The Canadian Encyclopedia. <https://www.thecanadianencyclopedia.ca/en/article/gitksan>.
- PySTAC Authors. (2017) 2025. *Stac-Utills/PyStac*. V. 1.12.2. November 8; Stac-utills, released. <https://github.com/stac-utills/pystac>.
- Radiantearth. 2024. "STAC: SpatioTemporal Asset Catalogs." <https://stacspect.org/en/>.
- Resource Inventory Standards Committee. 2001. *Reconnaissance (1:20 000) Fish and Fish Habitat Inventory Standards and Procedures*. <https://www2.gov.bc.ca/assets/gov/environment/natural-resource-stewardship/nr-laws-policy/risc/recce2c.pdf>.
- Roberge, M, J M B Hume, C K Minns, and T Slaney. 2002. *Life History Characteristics of Freshwater Fishes Occurring in British Columbia and the Yukon, with Major Emphasis on Stream Habitat Characteristics*. 262.
- Rosenfeld, Jordan, Marc Porter, and Eric Parkinson. 2000. *Habitat Factors Affecting the Abundance and Distribution of Juvenile Cutthroat Trout (Oncorhynchus Clarki) and Coho Salmon*

(*Oncorhynchus Kisutch*). 57: 9.

Schell, C. 2003. *A Brief Overview of Fish, Fisheries and Aquatic Habitat Resources in the Morice TSA*. <https://data.skeenasalmon.info/dataset/a-brief-overview-of-fish-fisheries-and-aquatic-habitat-resources-in-the-morice-tsa>.

Simoes, Rolf, Felipe Souza, Matheus Zaglia, Gilberto Ribeiro de Queiroz, Rafael dos Santos, and Karine Ferreira. 2021. "Rstac: An R Package to Access Spatiotemporal Asset Catalog Satellite Imagery." *2021 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 7674–77. <https://doi.org/10.1109/IGARSS47720.2021.9553518>.

Slaney, P A, and D Zaldokas. 1997. *Fish Habitat Rehabilitation Procedures*. https://www.for.gov.bc.ca/hfd/library/FFIP/Slaney_PA1997_A.pdf.

Sloat, Matthew R., Gordon H. Reeves, and Kelly R. Christiansen. 2017. "Stream Network Geomorphology Mediates Predicted Vulnerability of Anadromous Fish Habitat to Hydrologic Change in Southeast Alaska." *Global Change Biology* 23 (2): 604–20. <https://doi.org/10.1111/gcb.13466>.

stac-utils. 2025. *Stac-Utills/Stac-Fastapi-Pgstac: PostgreSQL Backend for Stac-Fastapi Using Pgstac* (<https://github.com/stac-utils/pgstac>). V. 4.0.2. Released. <https://github.com/stac-utils/stac-fastapi-pgstac>.

Stokes, J. 1956. *Upper Bulkley River Survey 1956*. <https://data.skeenasalmon.info/dataset/6d9cc7a6-683e-4de5-879e-77b592882a35/resource/9ff7c750-1a1d-40b3-9ee1-6bbb46bdf38e/download/dfo-ubulkley-hab-survey-1956.pdf>.

Swales, S., and C. D. Levings. 1989. "Role of Off-Channel Ponds in the Life Cycle of Coho Salmon (*Oncorhynchus Kisutch*) and Other Juvenile Salmonids in the Coldwater River, British Columbia." *Canadian Journal of Fisheries and Aquatic Sciences* 46 (2): 232–42. <https://doi.org/10.1139/f89-032>.

Tamblyn, Gregory C. 2005. *A Plan to Conserve and Protect Morice Watershed Fish Populations*. 78.

Thompson, Richard. 2013. *Assessing Fish Passage at Culverts – the Method, Its Metrics and Preliminary Findings from over 4,000 Assessments*. https://www2.gov.bc.ca/assets/gov/environment/plants-animals-and-ecosystems/fish-fish-habitat/fish-passage/assessing_fish_passage_at_culverts.pdf.

Thorley, J. L., S. Norris, and A. Irvine. 2021. *Channel Width 2021b*. <https://www.poissonconsulting.ca/f/859859031>.

Thorley, J, and A Irvine. 2021. "Channel Width 2021." <https://www.poissonconsulting.ca/f/1792764180>.

Toffolon, Marco, and Sebastiano Piccolroaz. 2015. "A Hybrid Model for River Water Temperature as a Function of Air Temperature and Discharge." *Environmental Research Letters* 10 (11): 114011. <https://doi.org/10.1088/1748-9326/10/11/114011>.

Washington Department of Fish & Wildlife. 2009. *Fish Passage Barrier and Surface Water Diversion Screening Assessment and Prioritization Manual*. Washington Department of Fish; Wildlife.

Olympia, Washington. <https://wdfw.wa.gov/sites/default/files/publications/00061/wdfw00061.pdf>.

Wilson, Tim, and Ken Rabnett. 2007. *Fish Passage Assessment of Highway 16 and CN Rail in the Bulkley Watershed*. 124. <https://data.skeenasalmon.info/dataset/fish-passage-assessment-highway-16-cn-rail-bulkley>.

Woll, Christine, David Albert, and Diane Whited. 2017. *Salmon Ecological Systems*. The Nature Conservancy. https://www.conservationgateway.org/ConservationByGeography/NorthAmerica/UnitedStates/alaska/sw/cpa/Documents/TNC_A_Salmon_Ecological_Systems_Model_Nushagak_Kvichak.pdf.

References

Note that this reference was not actually cited in the body of the report but rather in the `index.Rmd` file `yam1 headerj` under the `no_cite` key.

Session Info

Information about the computing environment is important for reproducibility. A summary of the computing environment is saved to `session_info.csv`, which can be viewed and downloaded from [here](#).

– Session info

```
## setting value
## version R version 4.5.2 (2025-10-31)
## os      macOS Tahoe 26.2
## system  aarch64, darwin20
## ui      X11
## language (EN)
## collate en_US.UTF-8
## ctype   en_US.UTF-8
## tz      America/Vancouver
## date    2026-08-04
## pandoc  3.9.0.2 @ /opt/homebrew/bin/ (via rmarkdown)
## quarto  1.4.553 @ /usr/local/bin/quarto
##
```

– Packages

## package	* version	date (UTC)	lib	source
## arrow	* 22.0.0.1	2025-12-23	[2]	CRAN (R 4.5.2)
## askpass	1.2.1	2024-10-04	[1]	CRAN (R 4.5.0)
## assertthat	0.2.1	2019-03-21	[2]	CRAN (R 4.5.0)
## base64enc	0.1-6	2026-02-02	[1]	CRAN (R 4.5.2)
## bit	4.6.0	2025-03-06	[1]	CRAN (R 4.5.0)
## bit64	4.8.0	2026-04-21	[1]	CRAN (R 4.5.2)
## blob	1.3.0	2026-01-14	[1]	CRAN (R 4.5.2)
## bookdown	* 0.46	2025-12-05	[2]	CRAN (R 4.5.2)
## bslib	0.10.0	2026-01-26	[1]	CRAN (R 4.5.2)
## cachem	1.1.0	2024-05-16	[1]	CRAN (R 4.5.0)
## chk	0.10.0	2025-01-24	[1]	CRAN (R 4.5.0)
## class	7.3-23	2025-01-01	[2]	CRAN (R 4.5.2)
## classInt	0.4-11	2025-01-08	[1]	CRAN (R 4.5.0)
## cli	3.6.6	2026-04-09	[1]	CRAN (R 4.5.2)
## codetools	0.2-20	2024-03-31	[2]	CRAN (R 4.5.2)
## crayon	1.5.3	2024-06-20	[1]	CRAN (R 4.5.0)
## crosstalk	1.2.2	2025-08-26	[1]	CRAN (R 4.5.0)
## curl	7.1.0	2026-04-22	[1]	CRAN (R 4.5.2)
## DBI	1.3.0	2026-02-25	[1]	CRAN (R 4.5.2)
## dbplyr	2.5.2	2026-02-13	[1]	CRAN (R 4.5.2)
## devtools	2.4.6	2025-10-03	[2]	CRAN (R 4.5.0)
## digest	0.6.39	2025-11-19	[1]	CRAN (R 4.5.2)

Session Info

```
## doParallel          1.0.17      2022-02-07 [2] CRAN (R 4.5.0)
## dplyr                * 1.2.1      2026-04-03 [1] CRAN (R 4.5.2)
## DT                   0.34.0      2025-09-02 [2] CRAN (R 4.5.0)
## e1071                1.7-17      2025-12-18 [1] CRAN (R 4.5.2)
## elevatr              0.99.1      2025-09-10 [2] CRAN (R 4.5.0)
## ellipsis             0.3.2       2021-04-29 [2] CRAN (R 4.5.0)
## evaluate             1.0.5       2025-08-27 [1] CRAN (R 4.5.0)
## farver               2.1.2       2024-05-13 [1] CRAN (R 4.5.0)
## fasstr               0.5.3       2024-09-27 [2] CRAN (R 4.5.0)
## fastmap              1.2.0       2024-05-15 [1] CRAN (R 4.5.0)
## fishbc               * 0.2.1.9000 2026-01-15 [2] Github (lucy-
schick/fishbc@0607ffd)
## forcats              * 1.0.1      2025-09-25 [2] CRAN (R 4.5.0)
## foreach              1.5.2       2022-02-02 [2] CRAN (R 4.5.0)
## fpr                  * 1.2.0      2026-02-03 [2] Github
(NewGraphEnvironment/fpr@2817533)
## fs                   2.1.0       2026-04-18 [1] CRAN (R 4.5.2)
## generics             0.1.4       2025-05-09 [1] CRAN (R 4.5.0)
## ggplot2              * 4.0.3      2026-04-22 [1] CRAN (R 4.5.2)
## gh                   1.5.0       2025-05-26 [2] CRAN (R 4.5.0)
## glue                 1.8.1       2026-04-17 [1] CRAN (R 4.5.2)
## gtable               0.3.6       2024-10-25 [1] CRAN (R 4.5.0)
## here                 1.0.2       2025-09-15 [2] CRAN (R 4.5.0)
## hms                  1.1.4       2025-10-17 [1] CRAN (R 4.5.0)
## htmltools            0.5.9       2025-12-04 [1] CRAN (R 4.5.2)
## htmlwidgets          1.6.4       2023-12-06 [1] CRAN (R 4.5.0)
## httpr                1.4.8       2026-02-13 [1] CRAN (R 4.5.2)
## iterators            1.0.14      2022-02-05 [2] CRAN (R 4.5.0)
## janitor              2.2.1       2024-12-22 [1] CRAN (R 4.5.0)
## jquerylib            0.1.4       2021-04-26 [1] CRAN (R 4.5.0)
## jsonlite             2.0.0       2025-03-27 [1] CRAN (R 4.5.0)
## kableExtra           * 1.4.0       2024-01-24 [2] CRAN (R 4.5.0)
## KernSmooth           2.23-26     2025-01-01 [2] CRAN (R 4.5.2)
## knitr                * 1.51        2025-12-20 [1] CRAN (R 4.5.2)
## labeling             0.4.3       2023-08-29 [1] CRAN (R 4.5.0)
## lattice              0.22-9      2026-02-09 [2] CRAN (R 4.5.2)
## leafem               * 0.2.5       2025-08-28 [2] CRAN (R 4.5.0)
## leaflet              * 2.2.3       2025-09-04 [1] CRAN (R 4.5.0)
## lifecycle            1.0.5       2026-01-08 [1] CRAN (R 4.5.2)
## lubridate            * 1.9.5       2026-02-04 [1] CRAN (R 4.5.2)
## magick               2.9.0       2025-09-08 [2] CRAN (R 4.5.0)
## magrittr             2.0.5       2026-04-04 [1] CRAN (R 4.5.2)
## memoise              2.0.1       2021-11-26 [1] CRAN (R 4.5.0)
## ngr                  * 0.0.1       2026-01-22 [2] Github
(newgraphenvironment/ngr@22c9714)
## otel                 0.2.0       2025-08-29 [1] CRAN (R 4.5.0)
## pagdown              * 0.23        2025-08-20 [2] CRAN (R 4.5.0)
```

```

## pak                0.11.1      2026-07-22 [1] CRAN (R 4.5.2)
## pdftools           * 3.7.0      2026-01-30 [2] CRAN (R 4.5.2)
## pillar             1.11.1      2025-09-17 [1] CRAN (R 4.5.0)
## pkgbuild           1.4.8        2025-05-26 [2] CRAN (R 4.5.0)
## pkgconfig          2.0.3        2019-09-22 [1] CRAN (R 4.5.0)
## pkgload            1.5.0        2026-02-03 [2] CRAN (R 4.5.2)
## plyr               1.8.9        2023-10-02 [1] CRAN (R 4.5.0)
## png                0.1-9        2026-03-15 [1] CRAN (R 4.5.2)
## prettyunits        1.2.0        2023-09-24 [1] CRAN (R 4.5.0)
## processx           3.9.0        2026-04-22 [1] CRAN (R 4.5.2)
## progress           1.2.3        2023-12-06 [2] CRAN (R 4.5.0)
## progressr          0.18.0       2025-11-06 [2] CRAN (R 4.5.0)
## proxy              0.4-29       2025-12-29 [1] CRAN (R 4.5.2)
## purrr              * 1.2.2        2026-04-10 [1] CRAN (R 4.5.2)
## qpdf               1.4.1        2025-07-02 [2] CRAN (R 4.5.0)
## R6                  2.6.1        2025-02-15 [1] CRAN (R 4.5.0)
## rappdirs           0.3.4        2026-01-17 [1] CRAN (R 4.5.2)
## raster             3.6-32       2025-03-28 [1] CRAN (R 4.5.0)
## rayshader          0.37.3       2024-02-21 [2] CRAN (R 4.5.0)
## rbbt                * 0.0.0.9000  2026-01-15 [2] Github
(paleolimbot/rbbt@212680c)
## RColorBrewer       1.1-3        2022-04-03 [1] CRAN (R 4.5.0)
## Rcpp               1.1.1-1.1    2026-04-24 [1] CRAN (R 4.5.2)
## RcppRoll           0.3.1        2024-07-07 [2] CRAN (R 4.5.0)
## readr              * 2.2.0        2026-02-19 [1] CRAN (R 4.5.2)
## readwritesqlite * 0.2.0.9021  2026-01-15 [2] Github
(poissonconsulting/readwritesqlite@ccdb297)
## remotes            2.5.0        2024-03-17 [2] CRAN (R 4.5.0)
## rgl                 1.3.31       2025-11-14 [2] CRAN (R 4.5.2)
## rlang              1.2.0        2026-04-06 [1] CRAN (R 4.5.2)
## rmarkdown          * 2.31        2026-03-26 [1] CRAN (R 4.5.2)
## RPostgres          * 1.4.10       2026-02-16 [1] CRAN (R 4.5.2)
## rprojroot          2.1.1        2025-08-26 [2] CRAN (R 4.5.0)
## RSQLite            3.52.0       2026-05-10 [1] CRAN (R 4.5.2)
## rstudioapi         0.18.0       2026-01-16 [2] CRAN (R 4.5.2)
## rvest              1.0.5        2025-08-29 [2] CRAN (R 4.5.0)
## s2                  1.1.9        2025-05-23 [1] CRAN (R 4.5.0)
## S7                  0.2.2        2026-04-22 [1] CRAN (R 4.5.2)
## sass                0.4.10       2025-04-11 [1] CRAN (R 4.5.0)
## scales             1.4.0        2025-04-24 [1] CRAN (R 4.5.0)
## sessioninfo        1.2.3        2025-02-05 [2] CRAN (R 4.5.0)
## sf                  * 1.1-0        2026-02-24 [1] CRAN (R 4.5.2)
## snakecase          0.11.1       2023-08-27 [1] CRAN (R 4.5.0)
## sp                  2.2-1        2026-02-13 [1] CRAN (R 4.5.2)
## staticimports      * 0.0.0.9001  2026-01-15 [2] Github
(newgraphenviroment/staticimports@19cd73c)
## stringi            1.8.7        2025-03-27 [1] CRAN (R 4.5.0)

```

Session Info

```
## stringr          * 1.6.0      2025-11-04 [1] CRAN (R 4.5.0)
## svglite          2.2.2      2025-10-21 [2] CRAN (R 4.5.0)
## systemfonts      1.3.1      2025-10-01 [2] CRAN (R 4.5.0)
## terra            1.9-11     2026-03-26 [1] CRAN (R 4.5.2)
## textshaping      1.0.4      2025-10-10 [2] CRAN (R 4.5.0)
## tibble           * 3.3.1      2026-01-11 [1] CRAN (R 4.5.2)
## tidyhydat        0.7.2      2025-10-23 [2] CRAN (R 4.5.0)
## tidyr            * 1.3.2      2025-12-19 [1] CRAN (R 4.5.2)
## tidyselect       1.2.1      2024-03-11 [1] CRAN (R 4.5.0)
## tidyverse        * 2.0.0      2023-02-22 [2] CRAN (R 4.5.0)
## tidyxl           1.0.10     2026-01-15 [2] Github
## (nacnudus/tidyxl@7e2fbe7)
## timechange       0.4.0      2026-01-29 [1] CRAN (R 4.5.2)
## tzdb             0.5.0      2025-03-15 [1] CRAN (R 4.5.0)
## units            1.0-1      2026-03-11 [1] CRAN (R 4.5.2)
## usethis          3.2.1      2025-09-06 [2] CRAN (R 4.5.0)
## vctrs            0.7.3      2026-04-11 [1] CRAN (R 4.5.2)
## viridisLite      0.4.3      2026-02-04 [1] CRAN (R 4.5.2)
## vroom            1.7.1      2026-03-31 [1] CRAN (R 4.5.2)
## withr            3.0.2      2024-10-28 [1] CRAN (R 4.5.0)
## wk               0.9.5      2025-12-18 [1] CRAN (R 4.5.2)
## xfun             0.57       2026-03-20 [1] CRAN (R 4.5.2)
## xml2             1.5.2      2026-01-17 [1] CRAN (R 4.5.2)
## yaml             2.3.12     2025-12-10 [1] CRAN (R 4.5.2)
##
## [1] /Users/airvine/Library/R/arm64/4.5/library
## [2] /Library/Frameworks/R.framework/Versions/4.5-
## arm64/Resources/library
## * — Packages attached to the search path.
##
##
```

Attachment - Phase 1 Data and Photos

Data and photos for all Phase 1 fish passage assessments are provided online at https://www.newgraphenvironment.com/fish_passage_skeena_2025_reporting/appendix—phase-1-fish-passage-assessment-data-and-photos.html.

Attachment - Habitat Assessment and Fish Sampling Data

All field data collected is available [here](#).

Habitat assessment data is available for download here https://github.com/NewGraphEnvironment/fish_passage_skeena_2025_reporting/blob/main/data/habitat_confirmations.xls.

Attachment - Bayesian analysis to map stream discharge and temperature causal effects pathways

Details of this analysis and subsequent outputs can be reviewed in the report [Spatial Stream Network Analysis of Nechako Watershed Stream Temperatures 2022b](#) (Hill et al. 2024). At the time of reporting, ongoing work regarding the project was tracked <https://github.com/poissonconsulting/fish-passage-22/issues> and <https://github.com/poissonconsulting/fish-passage-22b/issues>.